Introduction to
Unmanned Aerial System (UAS)

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UAS (Unmanned Aerial Systems)

A **UAS** (Unmanned Aircraft System) is a complete system consisting of an Unmanned Aerial Vehicle (**UAV**) and all associated components required for its operation.



This includes

- The Aircraft itself
- Ground Control Stations,
- Communication links
- Launch and Recovery equipment
- Other supporting hardware or software

Types of UAVs

Classification of UAVs can be based on various factors such as:

- Weight
- Range
- Altitude
- Purpose
- Propulsion Systems
- Configuration

Based on Weight, Range and Altitude

Based on Weight:

- **Nano/Micro UAVs:** Weigh less than **250 grams**
- **Micro/Small UAVs:** Weigh between 250 grams to **7 kg**
- **Small UAVs:** Weigh between 7 kg to 25 kg
- **Medium UAVs:** Weigh between 25 kg and 150 kg
- **Large UAVs:** Weigh more than **150 kilograms**

- Based on Range Altitude and Endurance:
  - LADP – Low Altitude and Deep Penetration
  - MALE – Medium Altitude and Long Endurance
  - HALE – High Altitude and Long Endurance







Definition of Low Altitude?

Based on Purpose

Reconnaissance and Surveillance UAVs: Used for monitoring and intelligence gathering.

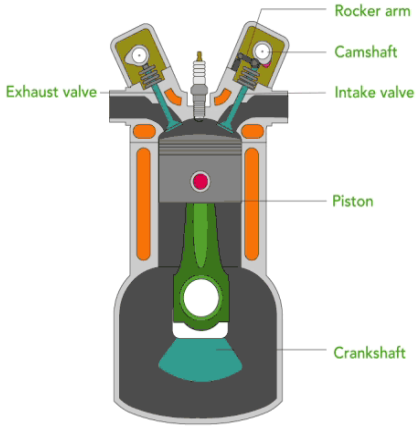
Combat UAVs: Armed drones for offensive military operations (e.g., UCAVs - Unmanned Combat Aerial Vehicles).

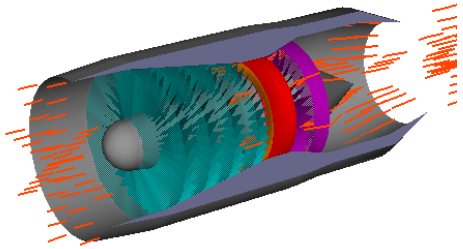
Logistics and Delivery UAVs: For transporting goods or supplies.

Research and Development UAVs: Used for testing and innovation.

Civilian UAVs: For photography, mapping, agriculture, etc.

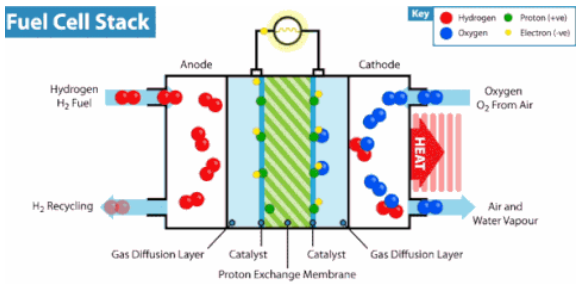
Based on Propulsion Type





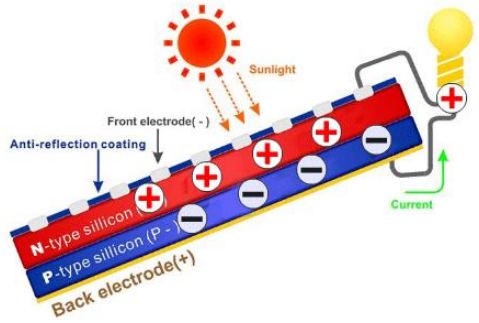
- Propulsion System
  - Hybrid
    - Combination Of Electrical & Mechanical(BLDC for VTOL and Engines for FW)
    - Combination of Mechnical and ···
  - Electrical
    - BLDC Motors
    - Brused DC Motors
  - Mechanical
    - IC Engines
    - Turbo jet Engines
  - Aerodynamic Power(Gliders)

Fuel type









Based on Configurations





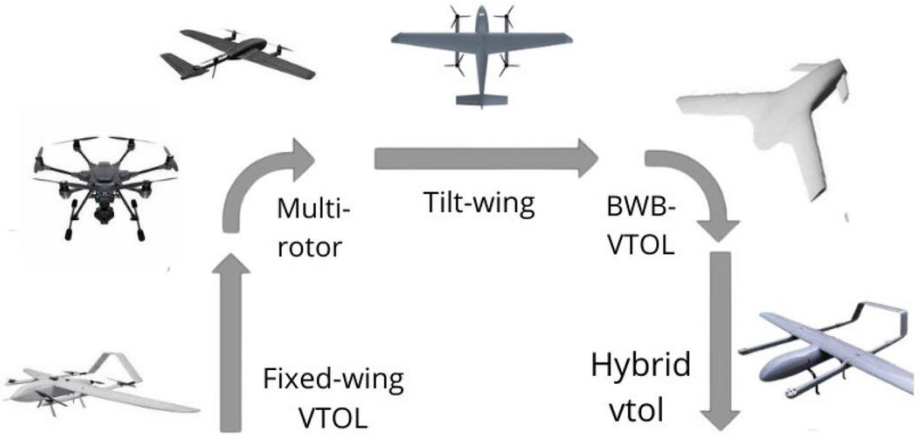




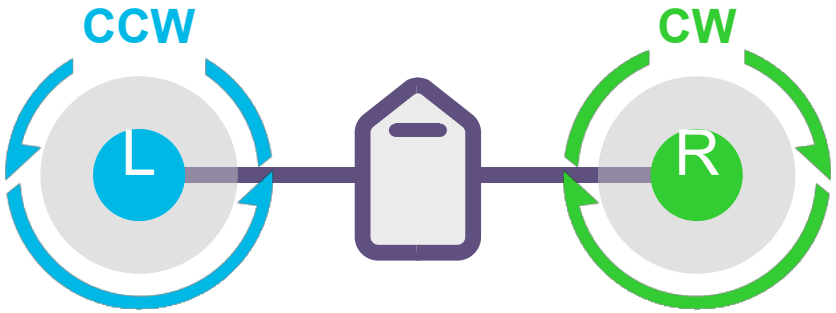
- Confidurations
  - Rotors
    - Single Rotor / Helicopters
    - Multirotor
      - Bi Copter
      - TriCopter
      - Quadcopters
        - X Type/H Type/ Plus Type
      - Hexacopters
      - Octocopters
        - CoAxial
    - Conventional Type
    - Flying wing/Delta
    - V Tail/T Tail/Inverted V/H Tail
  - Fixed Wing
  - Hybrid
    - VTOL Capability
    - Tail Sitter
    - Tilt Rotors
  - (Gliders)



Vertical **T**ake**O**ff and **L**anding (**VTOL**)



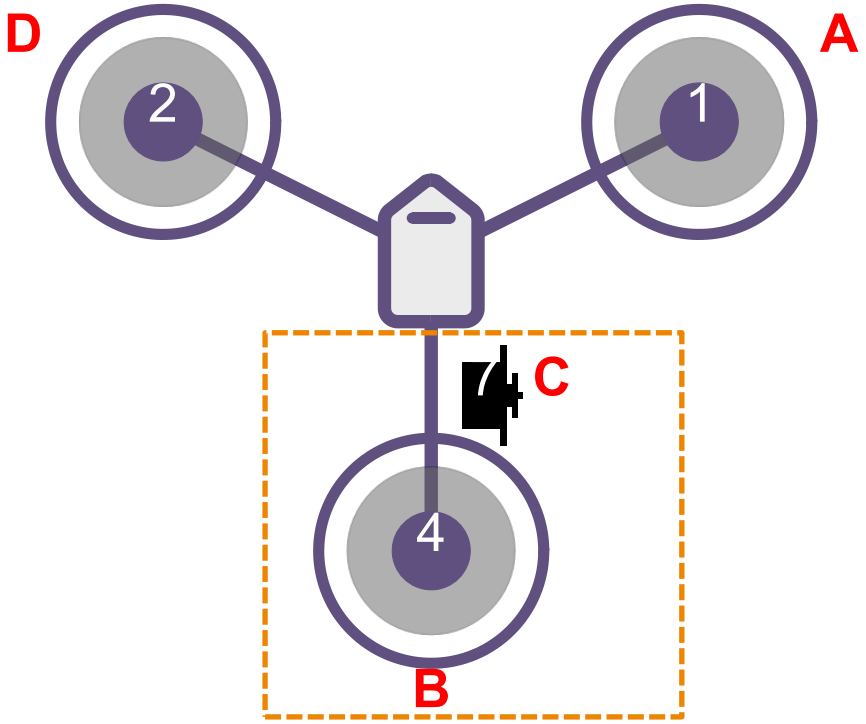
BiCopter and TriCopter:



BICOPTER

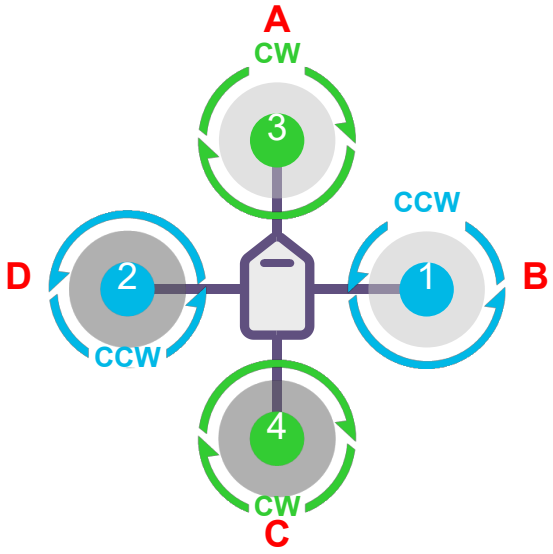
L = Left Throttle, Left Tilt

R = Right Throttle, Right Tilt

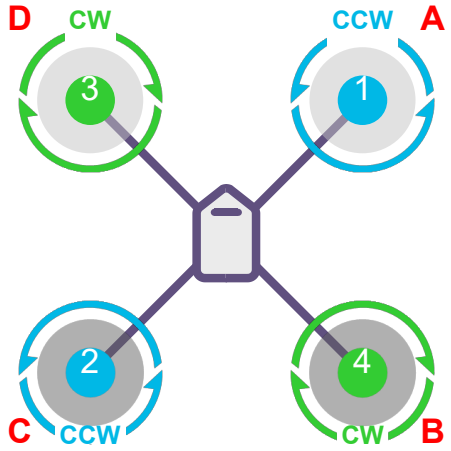


TRICOPTER

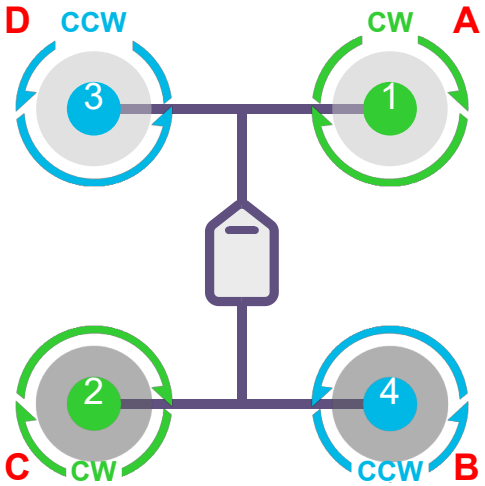
QuadCopter



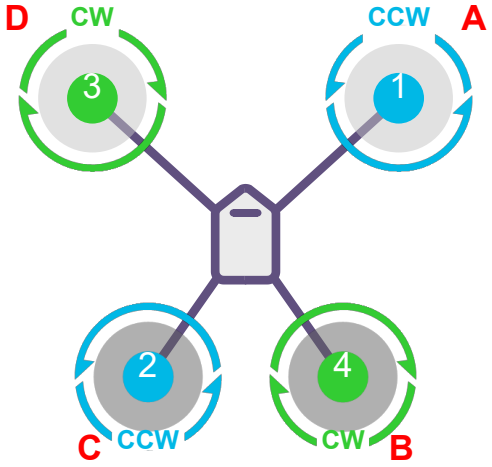
QUAD PLUS



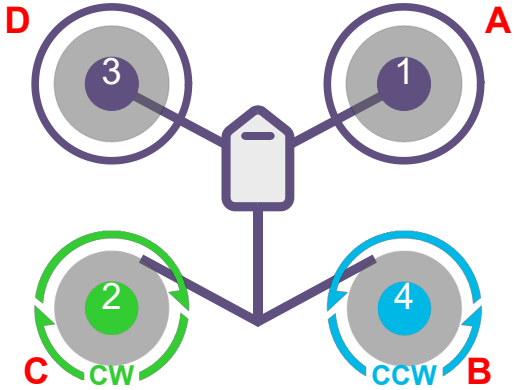
QUAD X



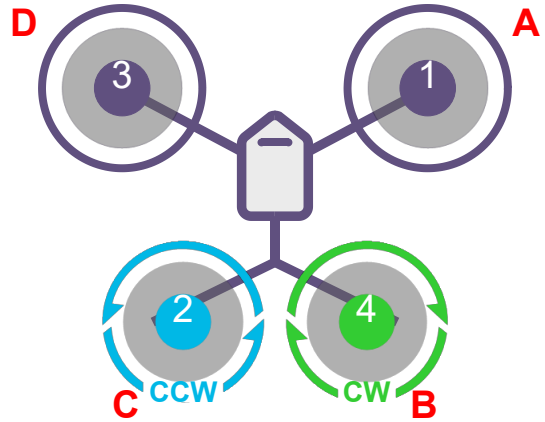
QUAD H



QUAD V



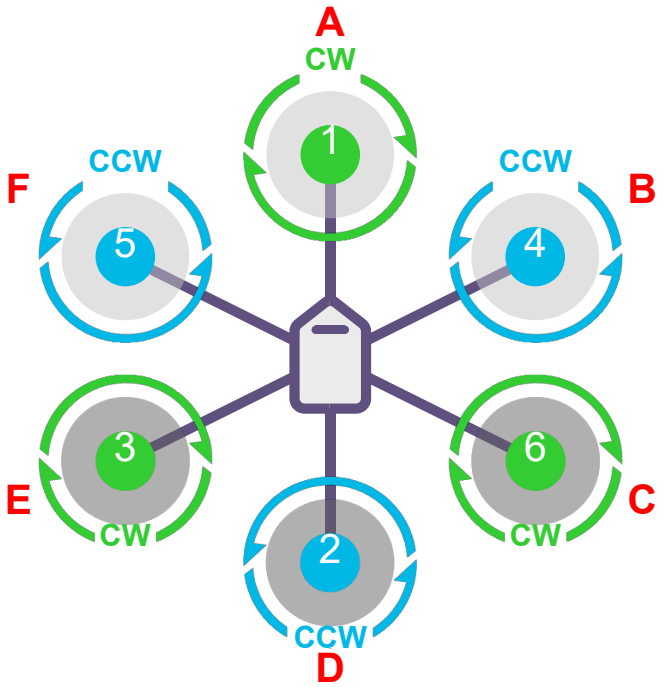
QUAD V TAIL



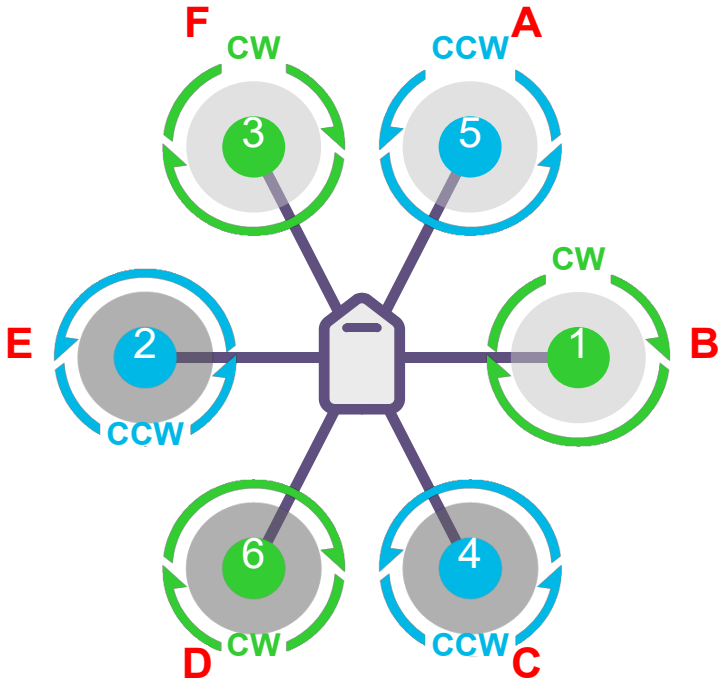
QUAD A TAIL



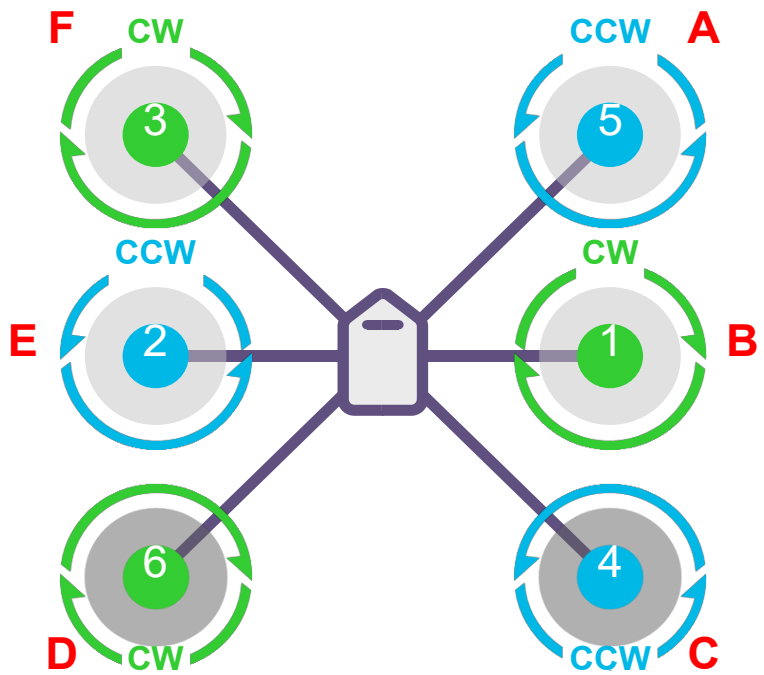
HexaCopter



HEXA PLUS

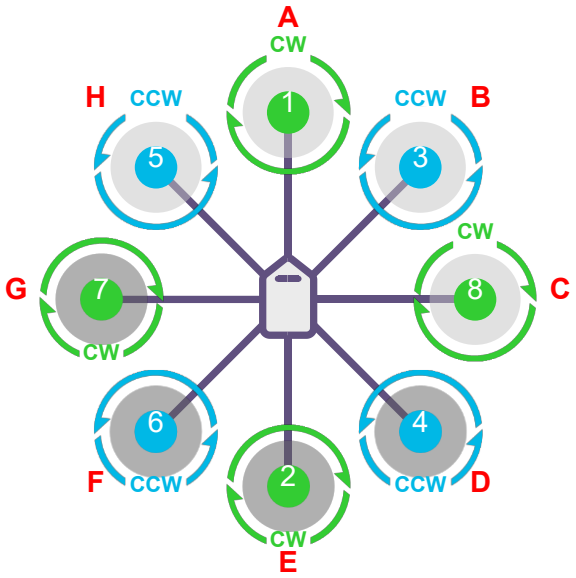


HEXA X

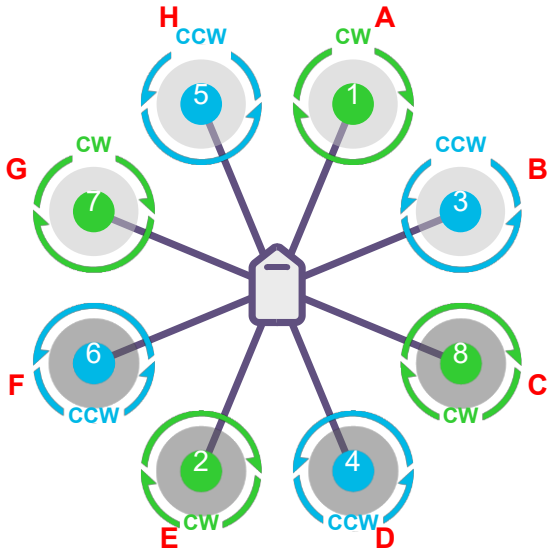


HEXA H

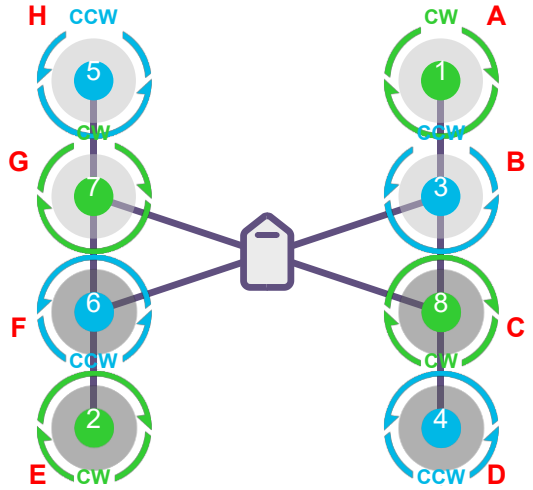
OctoCopter



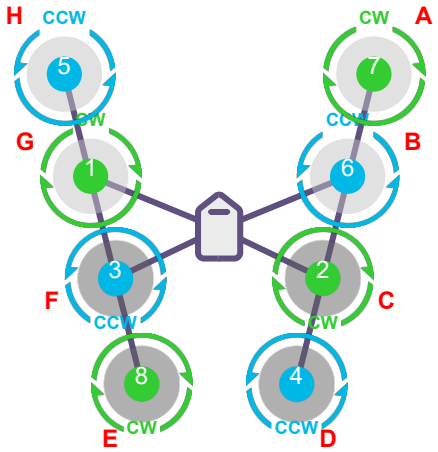
OCTO PLUS



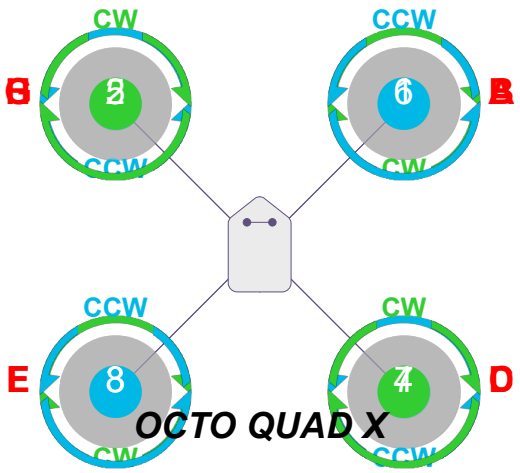
OCTO X

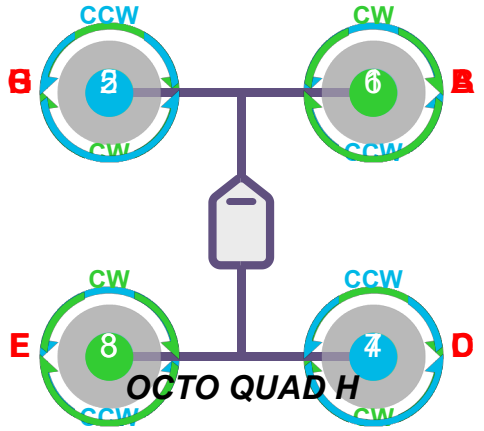


OCTO H



OCTO V





Based on Level of Autonomy





Autonomy Level	Decision Making	Human Role	System Capabilities
Level 1: Human-Operated	Human makes all decisions	Full control	Basic remote control functions
Level 2: Human-Delegated	Human delegates specific tasks	Task activation/deactivation	Automated controls, engine management
Level 3: Human-Supervised (HMI)	System operates with human oversight	Supervision and guidance	Task execution based on sensor data
Level 4: Fully Autonomous (Ideal)	System makes operational decisions	High-level objective setting	Independent task execution, emergency handling

Conventional/unconventional UAVs



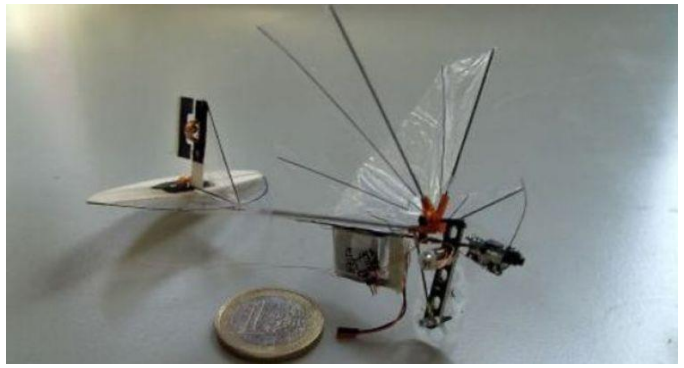


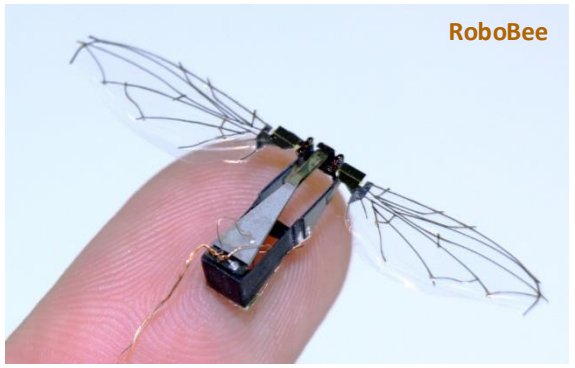












Ornithopters / Flapping-wing drones / Flapper drones

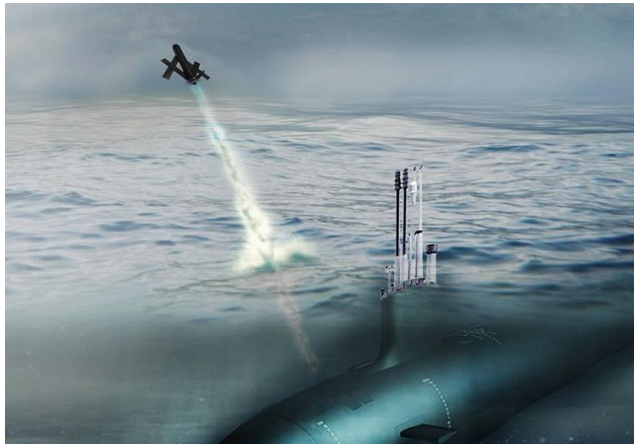








Conventional/unconventional UAVs



Conceptual 概念型

Blackwing · US Navy



Surface 水面型

SUWAVE · Université de Sherbrooke



Fixed Wing 固定翼型

AquaMAV · Imperial College London



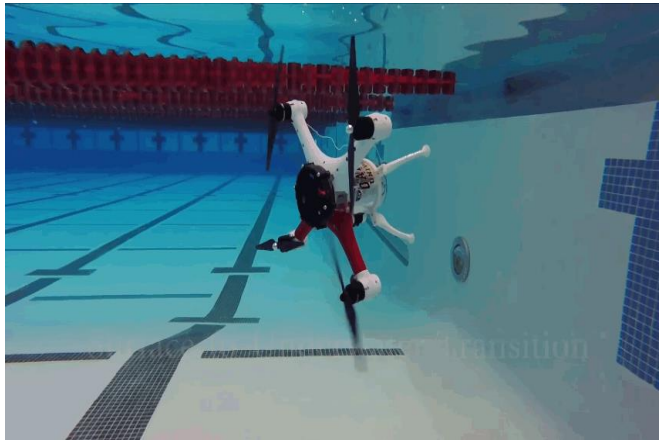
Quadrotor 四旋翼型

Naviator · Rutgers University



Hybrid 混合型

Nezha · Shanghai Jiaotong University



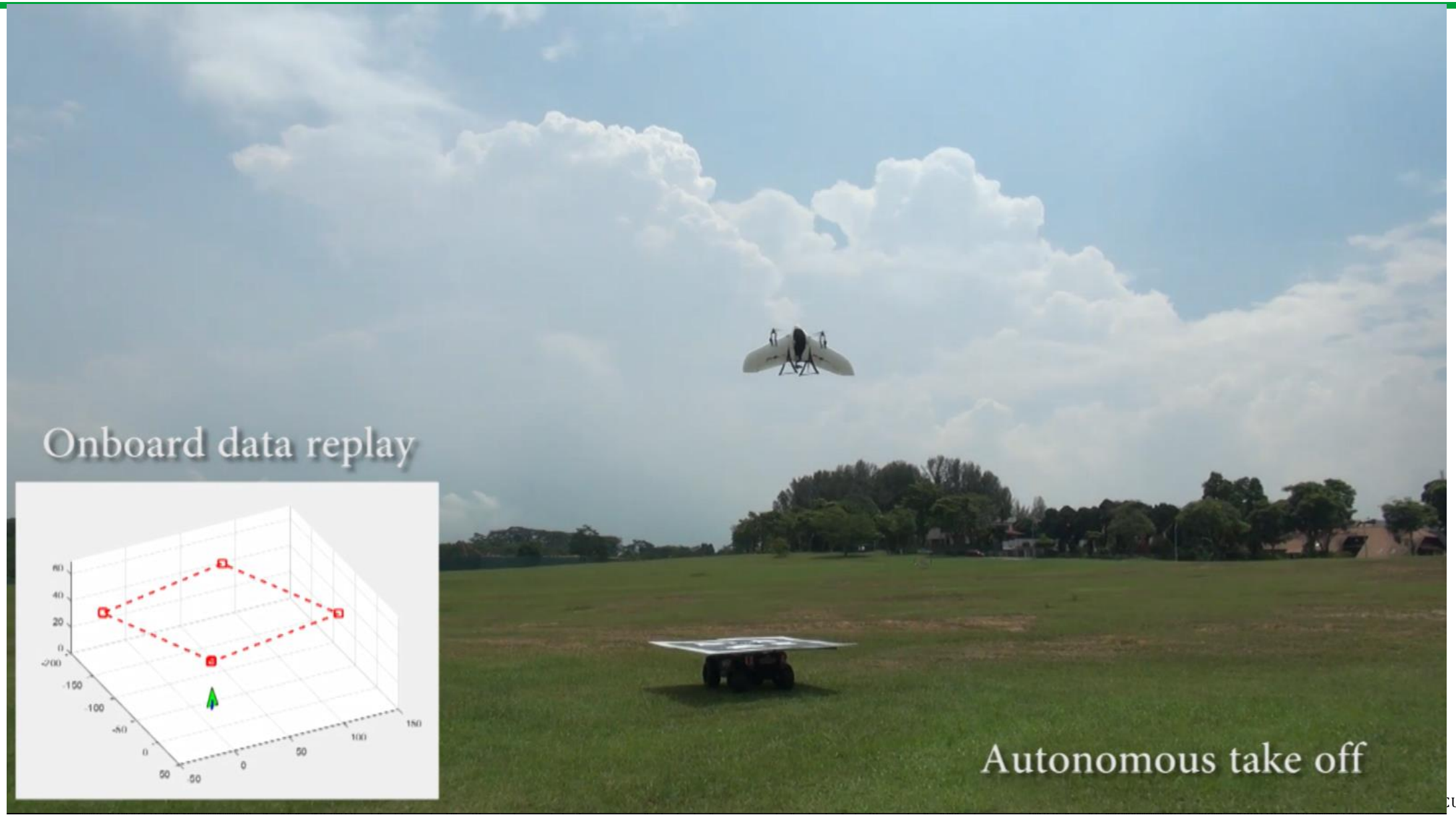
Quadrotor 四旋翼型

Loon Copter · Oakland University

Example: VTOL fixed-wing UAV prototype 1

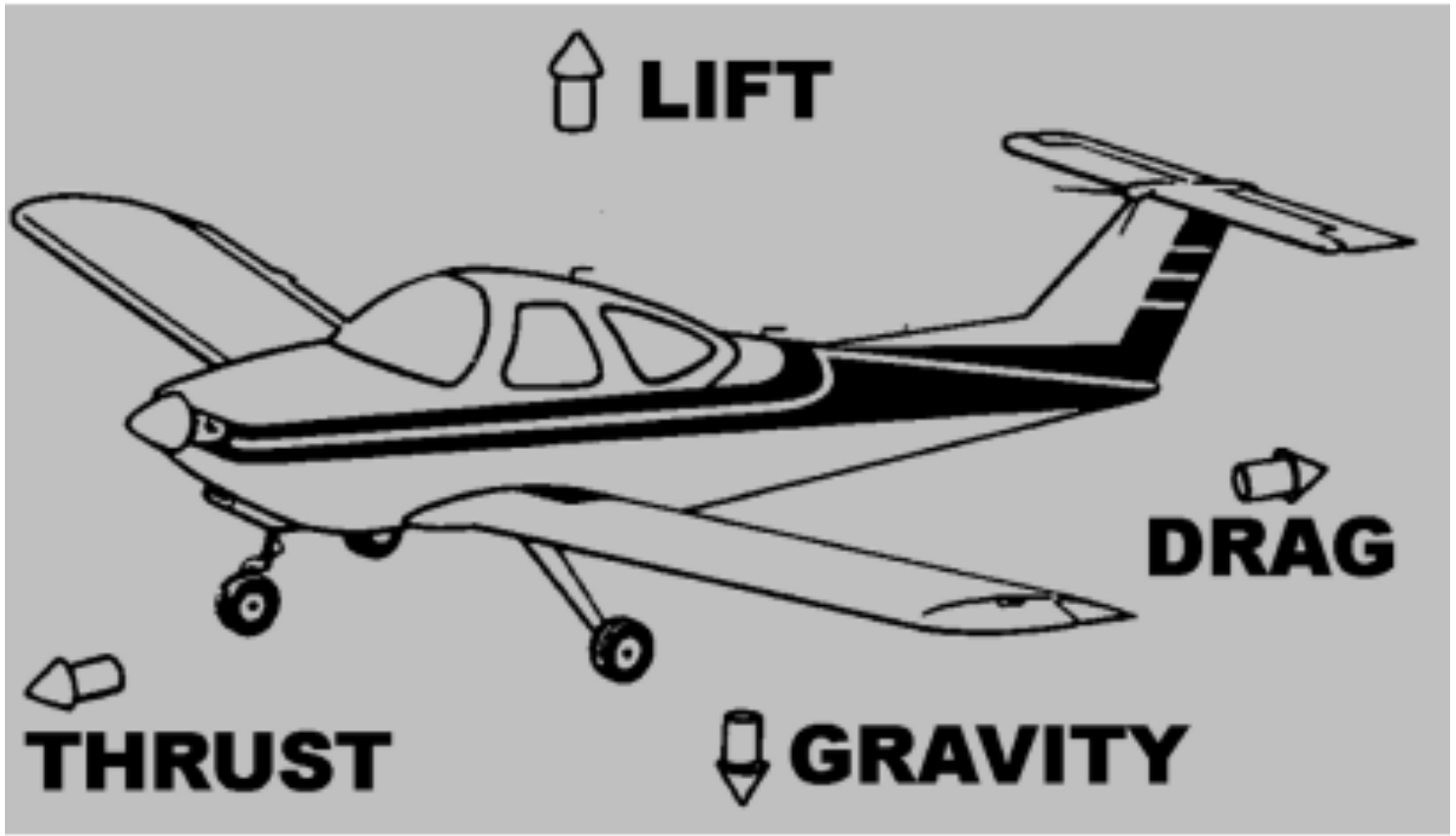


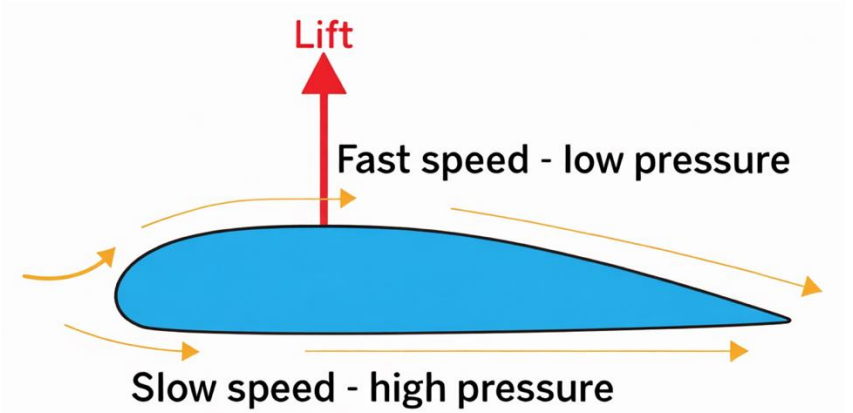
Example: VTOL fixed-wing UAV prototype 2



Basic principles of flight

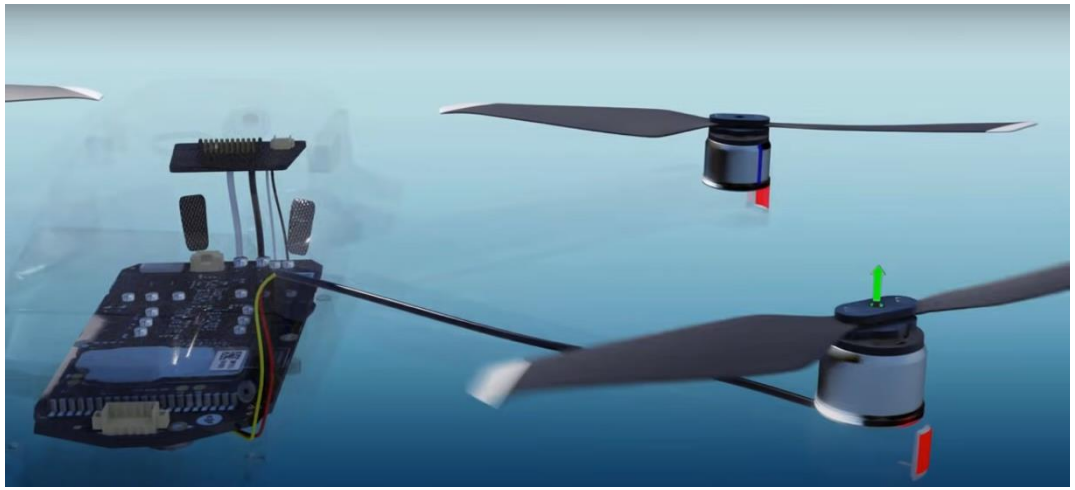
What principle of fluid dynamics?



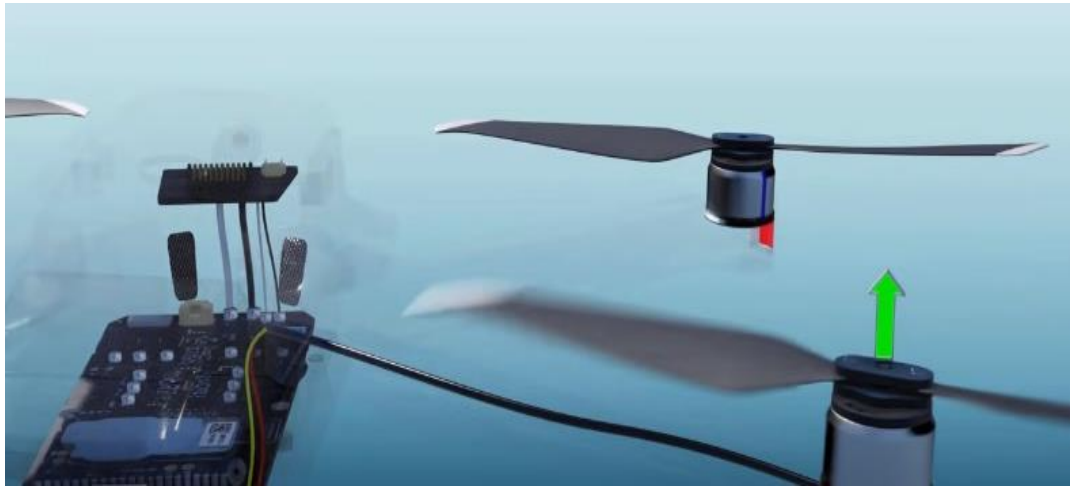




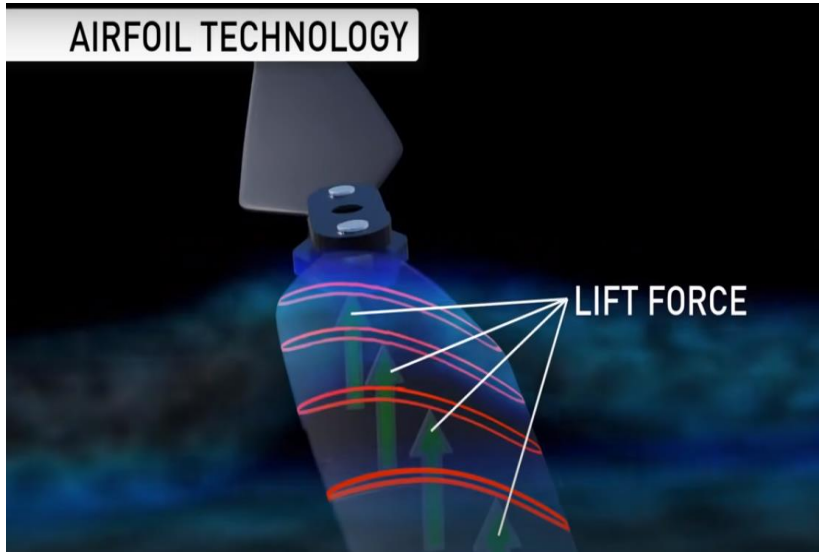
Flight principle for quadcopters



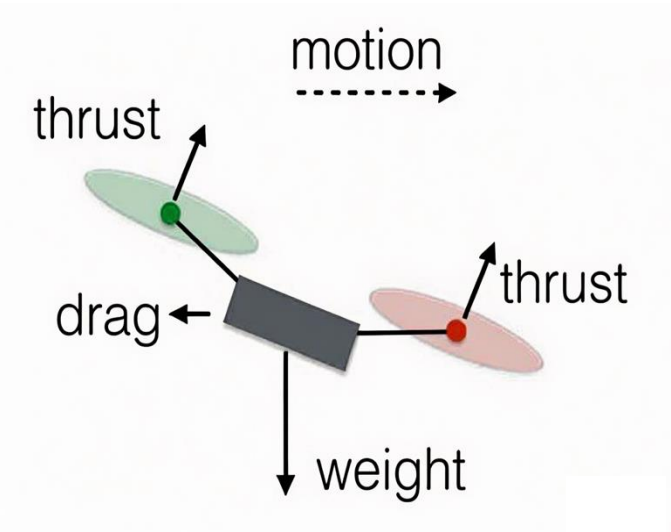
Blade Speed

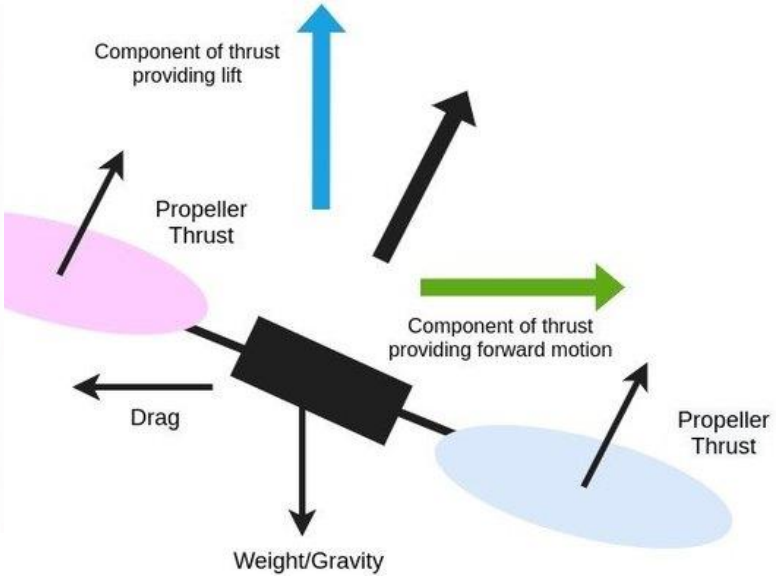


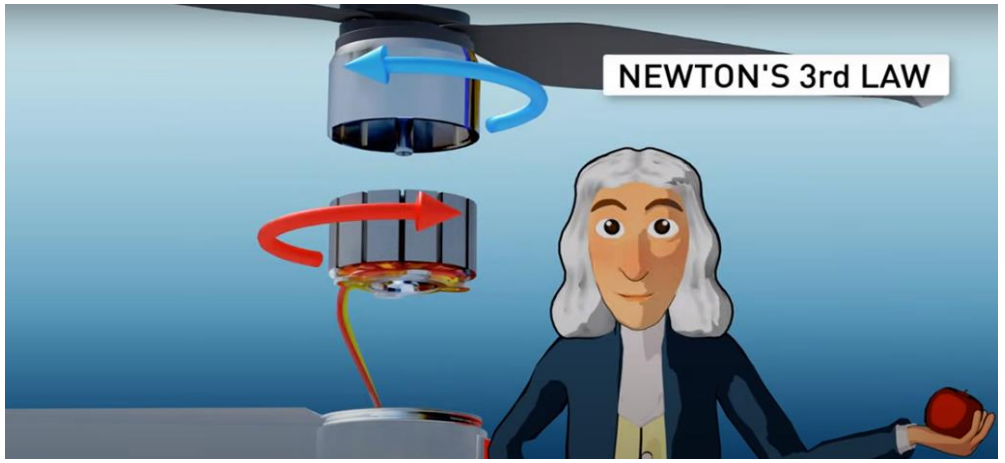
Lift Force



Trust & drag

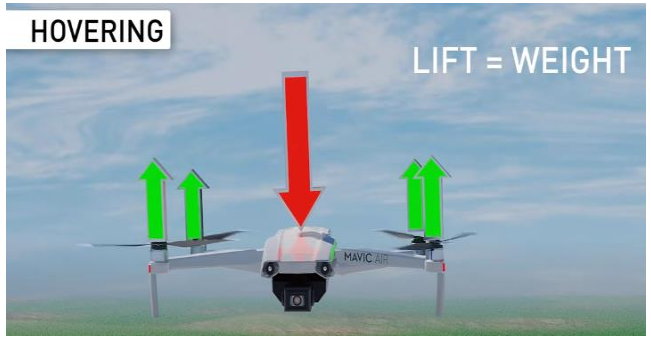


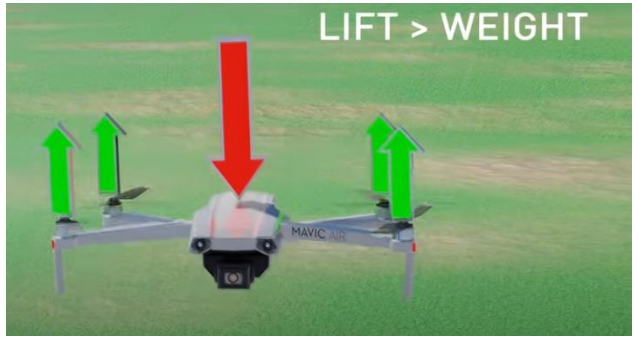


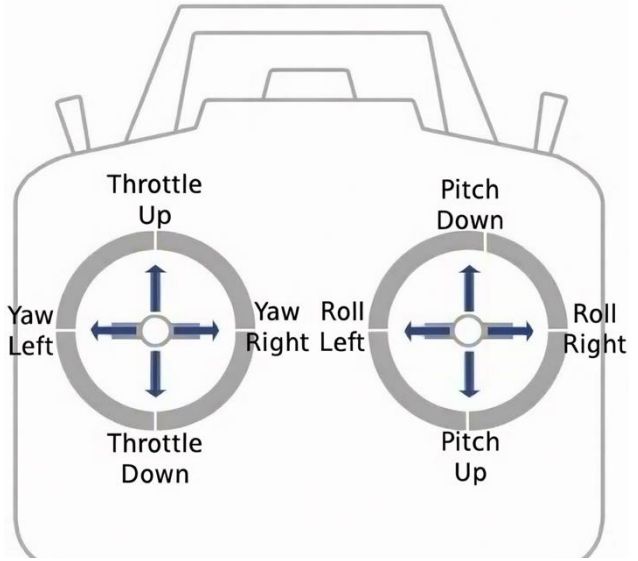


What will happen if all propellers rotate in the same directions???

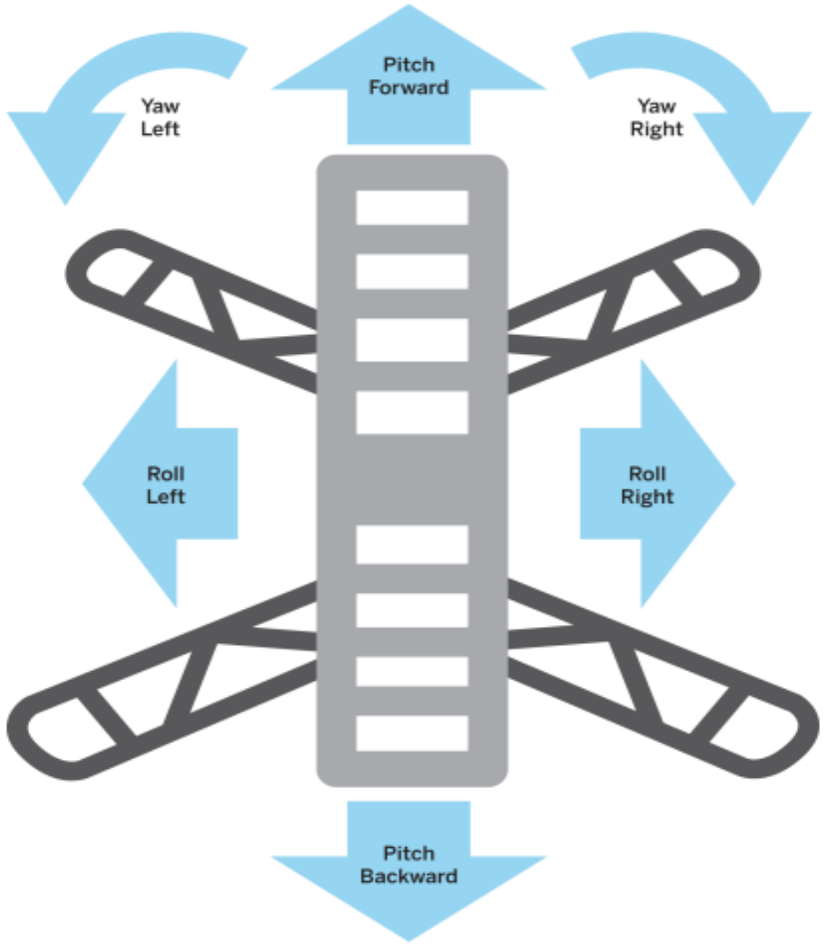
Throttle must generate enough lift to counter the Effects of Weight

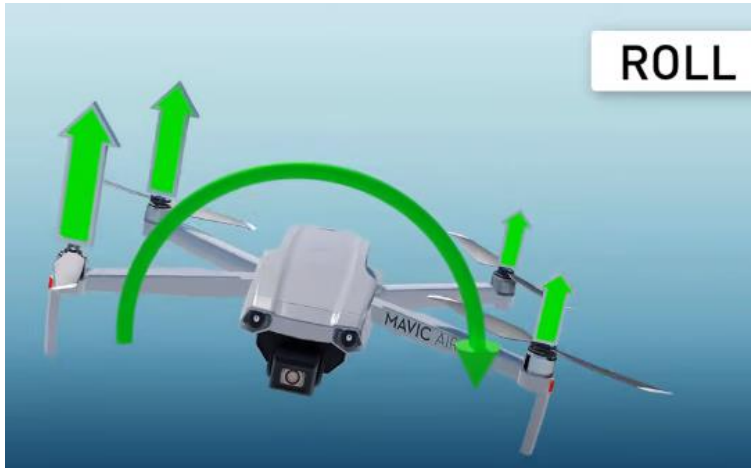






Yaw/Pitch/Roll



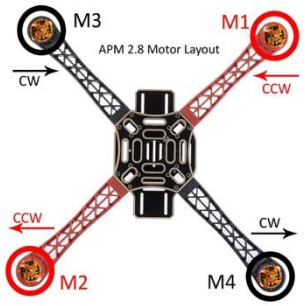


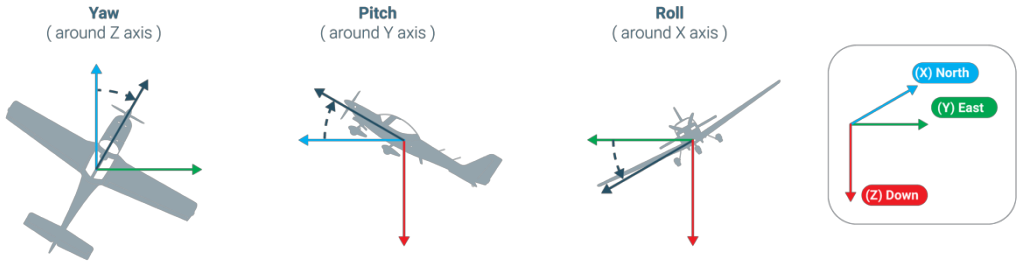


Dynamics of a Multicopter

Motor Rotation:

- Two motors rotate clockwise (CW), and two rotate counterclockwise (CCW) to ensure torque balance and maintain stable flight.
- By having two motors rotate clockwise and two counterclockwise, the torques cancel each other out, ensuring the quadcopter remains stable.



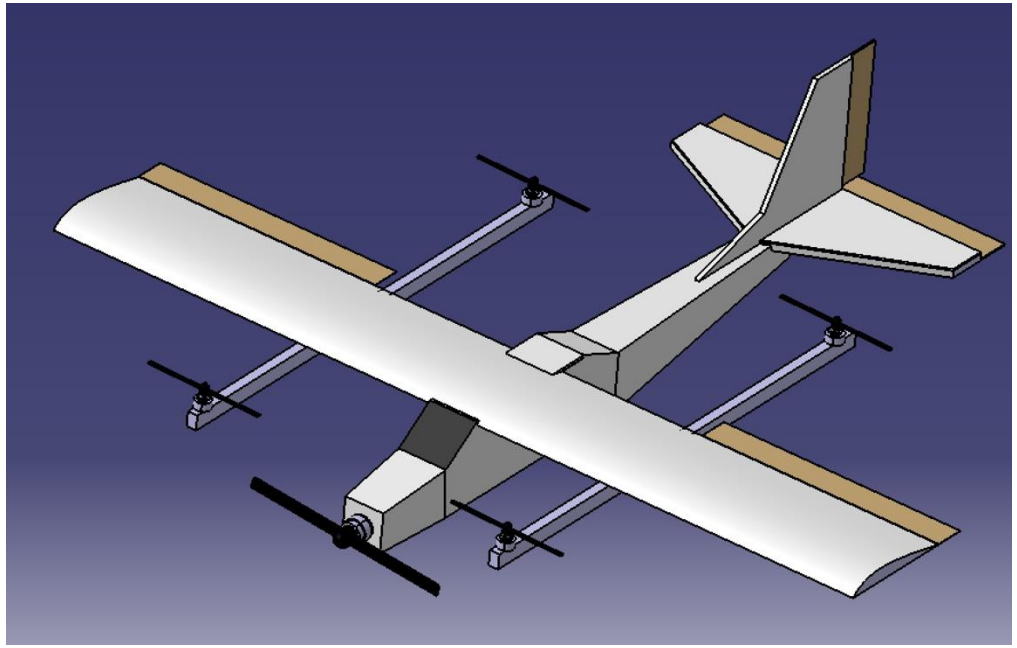


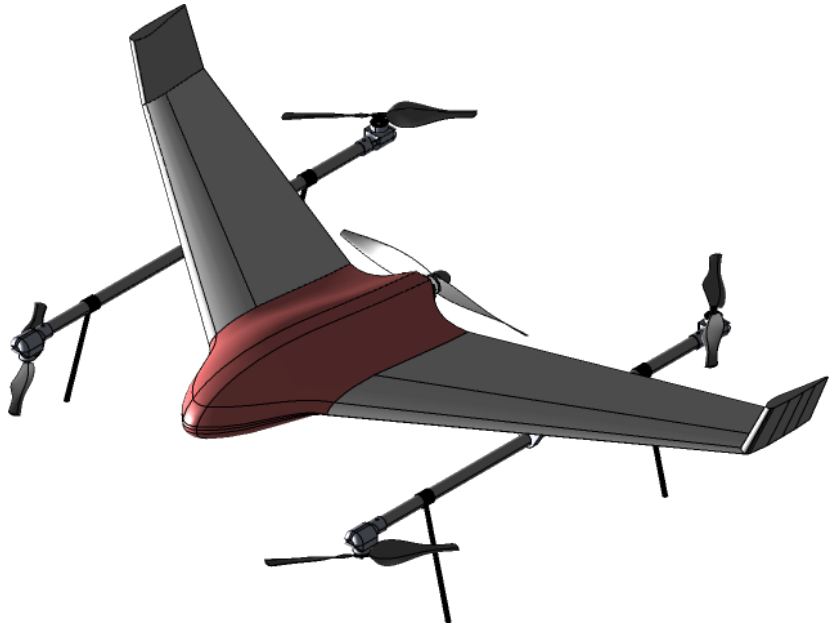
Summary of Control Actions:

Motion
Ascend
Descend
Roll
Pitch
Yaw

Dynamics of a hybrid VTOL

- Ascend/ Descend: The take-off and landing will be done with all the four motors rotating.
- Transition(Q2F, F2Q): During Q2F transition, the rear motors and the FW motor will rotate much faster; During landing, the front motors will rotate faster and the FW motor gradually stops.
- In Flight (Roll, Pitch, Yaw): All the **Control Surfaces** will take care of this.





Dynamics of a Tail-Sitter

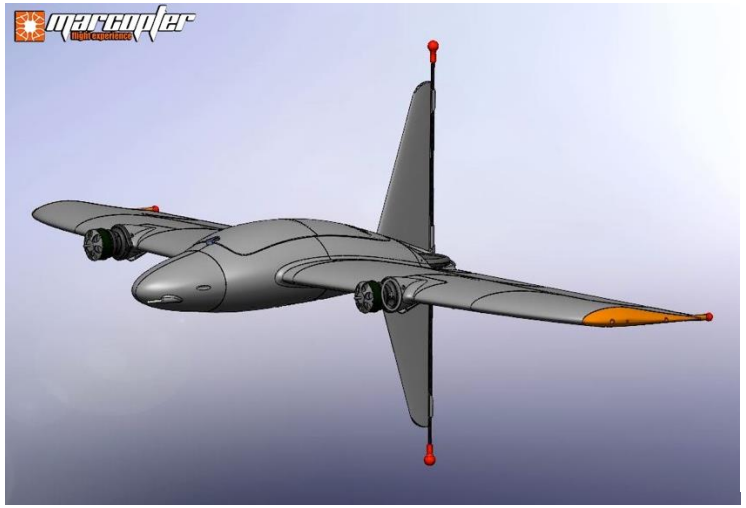
Ascend/ Descend: The aircraft starts in a vertical position, with its tail on the ground. Control surfaces or vectoring mechanisms adjust yaw, pitch, and roll during take-off.

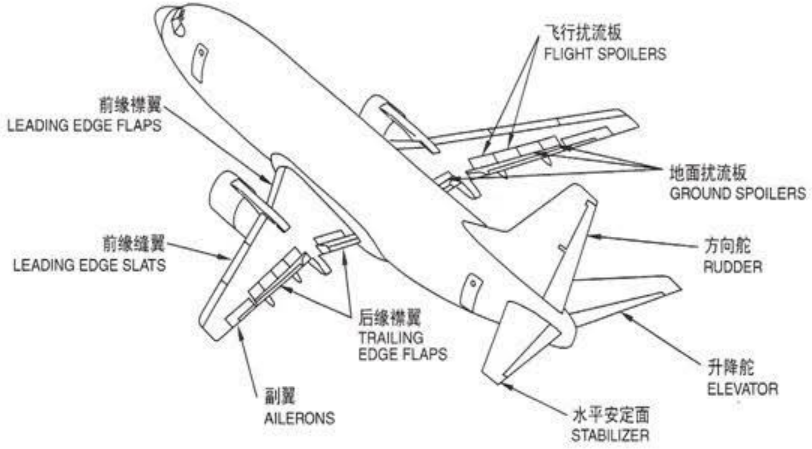
Transition (Forward): The aircraft begins tilting forward using a combination of aerodynamic control surfaces (e.g., elevons, ailerons) and motor thrust vectoring. This phase involves careful balance to avoid instability as lift transitions from propeller thrust to aerodynamic surfaces.

Transition (Reverse): The aircraft slows down by reducing forward thrust and using control surfaces to pitch up gradually. The aircraft transitions back to a vertical orientation for hovering.

In Flight (Roll, Pitch, Yaw? How?): All the **control surfaces** will take care of this.





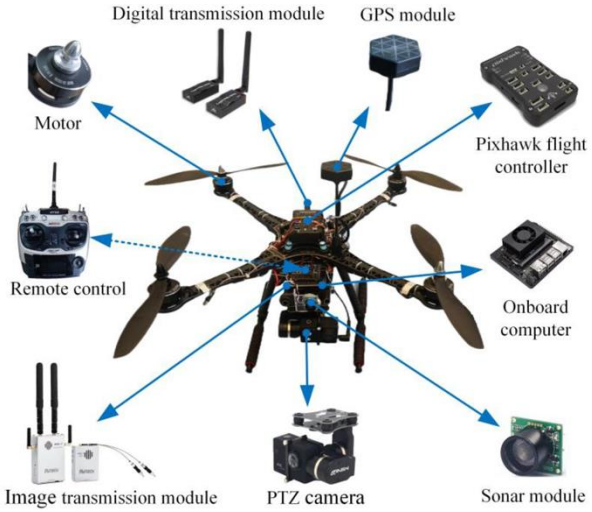


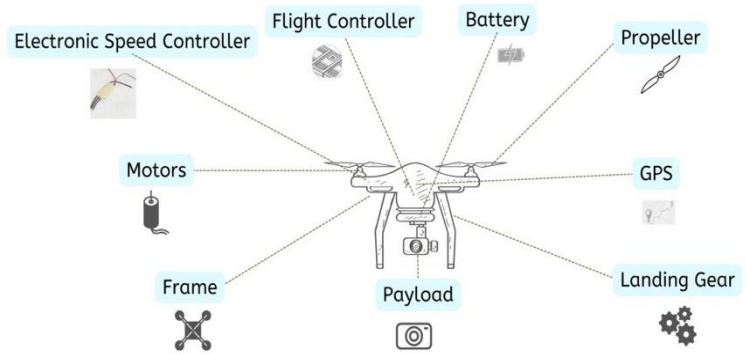
Components of UAVs

A UAV consists of several essential parts that work together to enable stable flight and control.

Here are the key component:

- Frame
- Motors
- Propellers
- Electronic Speed Controllers
- Flight Controller
- Power Distribution Board (PDB)
- Servos
- Battery
- IMU & GPS
- Other sensors (Ultrasonic & Flow Sensor, Lidar, Cameras – IR, RGB, Multispectral…)
- Transmitter and Receiver
- Camera Gimbal
- Ground control station (GCS) & Telemetry
- Landing Gear
- …





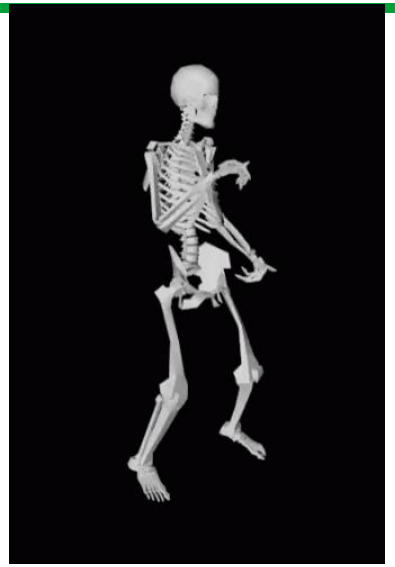


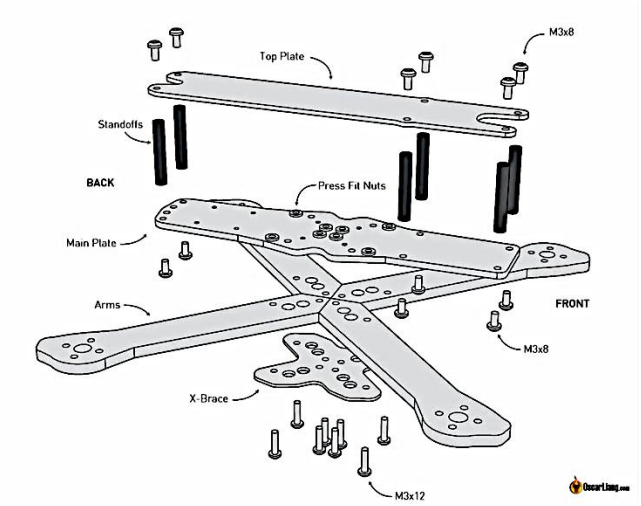


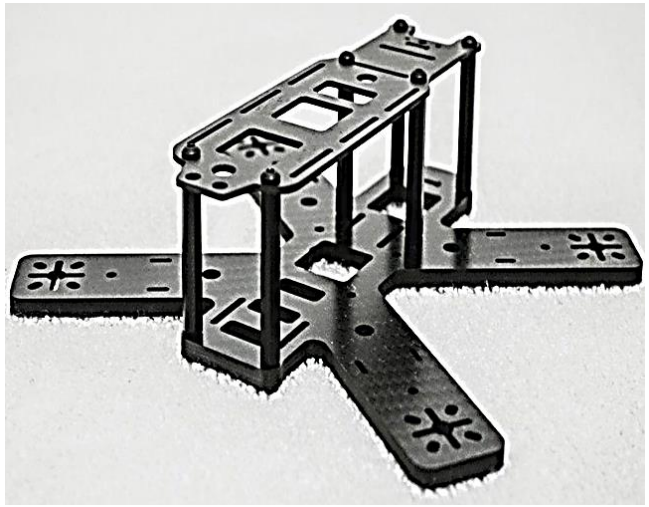
Frame (Airframe)

Structural backbone of a drone. It is the skeleton on which all other components are mounted, and provides protection to all the electronics inside the drone:

- The body houses and protects your electronic components including flight controller, ESC, camera, battery etc.
- Typically it consists of a bottom plate, top plate and some standoffs in between to hold them together.
- The arms are for installing the motors on.







Static airframe



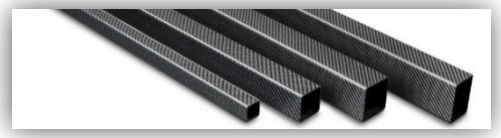
Foldable airframe

Frame selection

Shapes

- Tube shaped
- U-shaped
- T-shaped



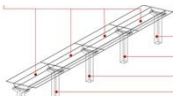


Structure

- Framed structure
- Simple trusses
- Diagonal brace
- Girder construction



















Material	Carbon Fiber	Fiberglass	Poly-carbonate	Acrylic	Aluminium Alloy	Wood
Density (kg/m³)	1750	2400	1200	1150	2700	300-600
Stiffness (MPsi)	9.3	2.7	0.75	0.38	10.3	0.16-0.9
Tensile Strength (KPsi)	120	15-50	8-16	8-11	15-75	1-4.6
Cost (1-10) 10 being cheapest	1	6	9	9	7	10
Manufacturing (1-10) 10 being easiest	3	7	6	7	7	10

- Quadcopter frame **size** (a.k.a. **wheelbase)** is the diagonal motor to motor distance measured in millimeters.
- When people say, “a five-inch frame,” it means the frame is sized for 5″ props.
- Use the biggest propellers a frame can support to maximize its performance.
- If you run 4-inch props on a 5″ frame, it would not perform as well as on a 4″ frame.



Frame – landing skid

- To support weight of UAV
- To ensure safe distance between propellers and the ground during landing
- Reduce **ground effect?** during take-off/landing
- Absorb impact upon landing



Ehang Ghost



DJI Flame Wheel F450



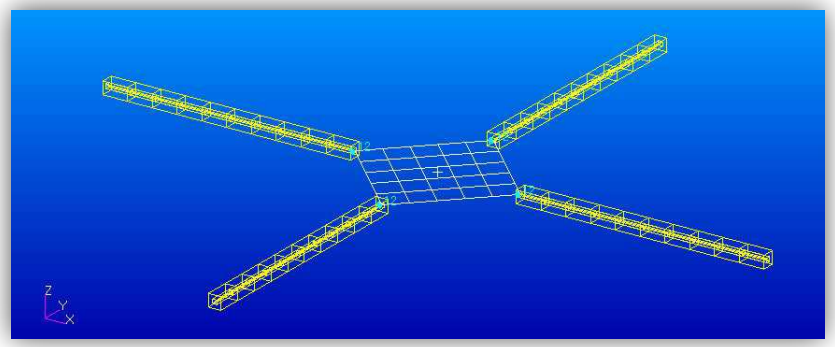
DJI Phantom

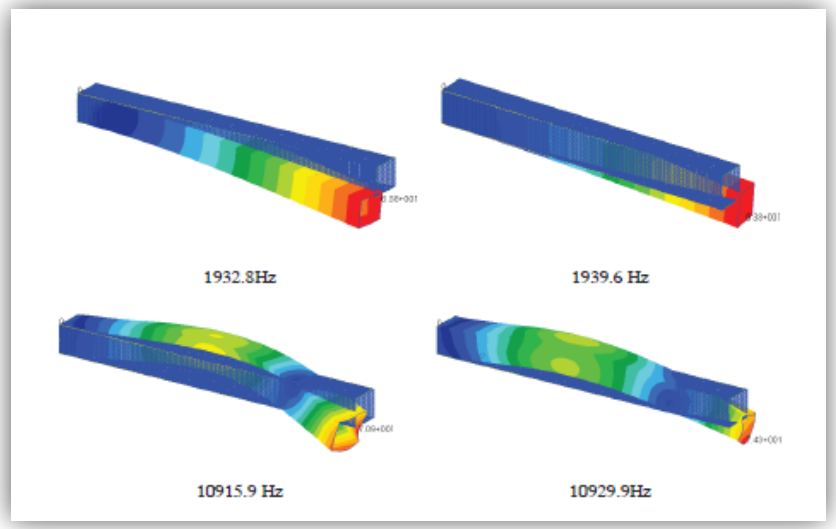
Frame tests

Finite Element Analysis (FEA)

Software

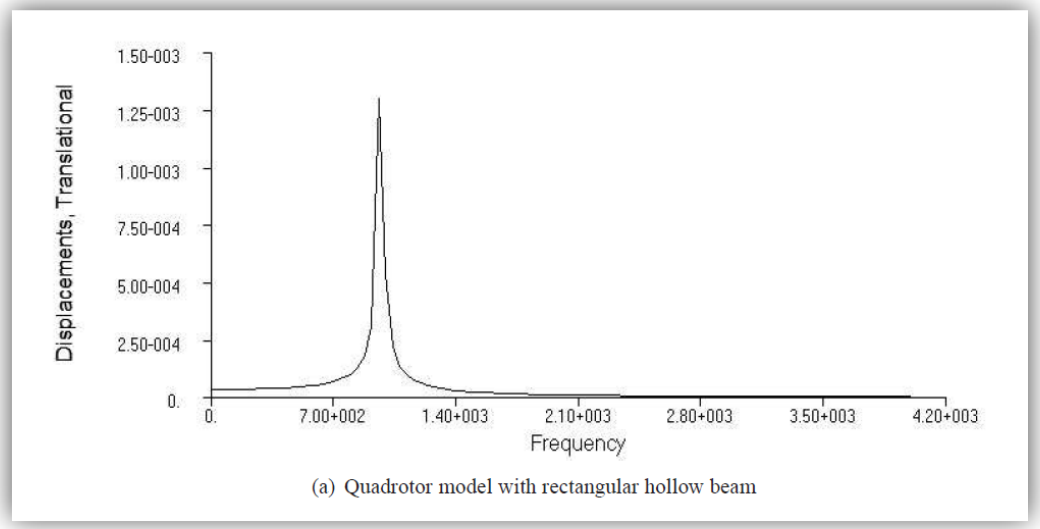
- MSC Patran
- MSC Nastran
- SolidWorks





What it does?

- Model structure of different shapes and assign corresponding material properties
- Analyze vibration frequency and amplitude



(a) Quadrotor model with rectangular hollow beam

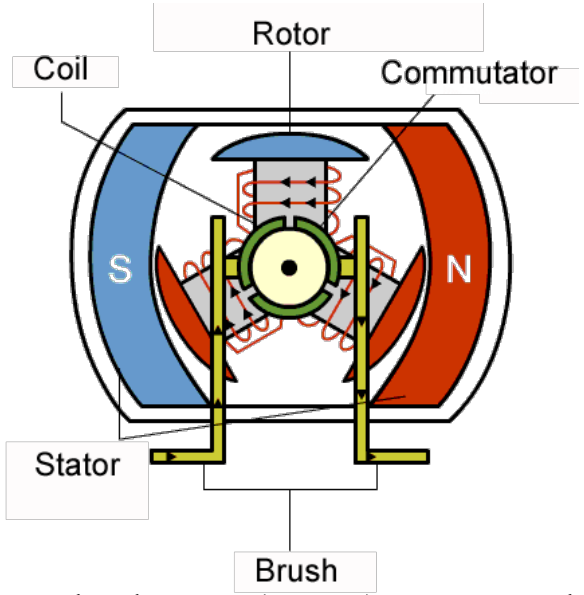
Motor selection

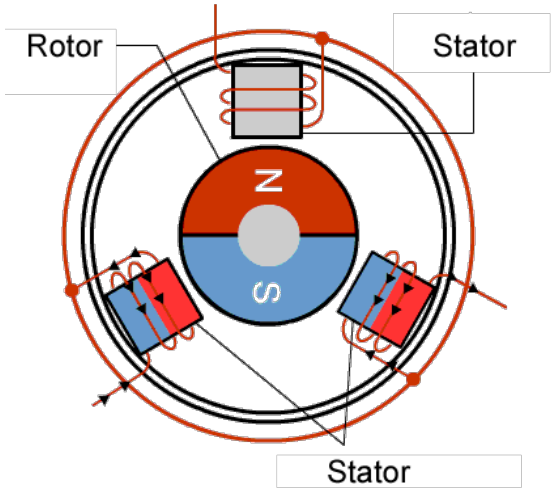
- Thrust-to-Weight Ratio: Motors should produce a total thrust of at least **2 times** of the quadcopter's weight (including frame, battery, and payload).
- KV Rating (how fast the motor rotates for a given voltage, example, 1000 KV is 1000 RPM per volt supplied to the motor):
  - Low KV motors (<1000 KV): Suitable for large propellers and heavy-lift quadcopters (more torque, lower RPM).
  - High KV motors (>1000 KV): Suitable for smaller propellers and lightweight racing drones (higher RPM, less torque).
- Motor Current Draw: Ensure the motor's current draw matches the ESC and battery capacity.
- Voltage: Select motors compatible with the battery voltage (e.g., 3S = 11.1V or 4S = 14.8V).



First 2-digit of the motor label corresponds to motor diameter in mm, last 2-digit of the label corresponds to height of the rotor in mm

BLDC Motor







- With a Brushed DC (BDC) motor, there is a brush which is connected with the commutator.
- Whenever it comes **in contact with the commutator**, it energises the electric coil and makes it as magnetic coil, and it repels from the permanent magnets.

- With a BL (brushless) DC motor, it is the **permanent magnet** that rotates.
- Rotation is achieved by **changing the direction of the magnetic fields** generated by the surrounding stationary coils.
- To control the rotation, you adjust the **magnitude and direction of the current** into these coils.

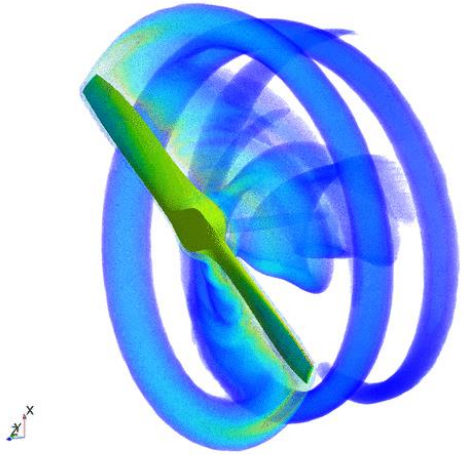

www.LearnEngineering.org

Propeller selection

- Size: Larger propellers (e.g., 10 inches) generate more thrust and are suitable for heavy quad copters; Smaller propellers (e.g., 5 inches) are better for racing and agility.
- Pitch: Higher pitch propellers (e.g., 4.5 inches) are faster but less efficient; Lower pitch propellers provide better stability and efficiency.
- Match the propeller size with the motor's recommended specifications to avoid overloading the motor.

A propeller can have a 9.4×4.3×2.

- 9.4, relates to the size, mostly in inches.
- 4.3, the second number, indicates pitch and is also in inches.
- 2 is the number of blades, the figure on the left shows a 2-bladed propeller.

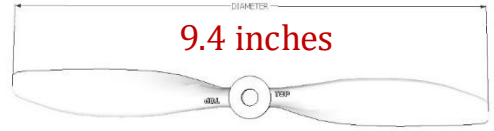




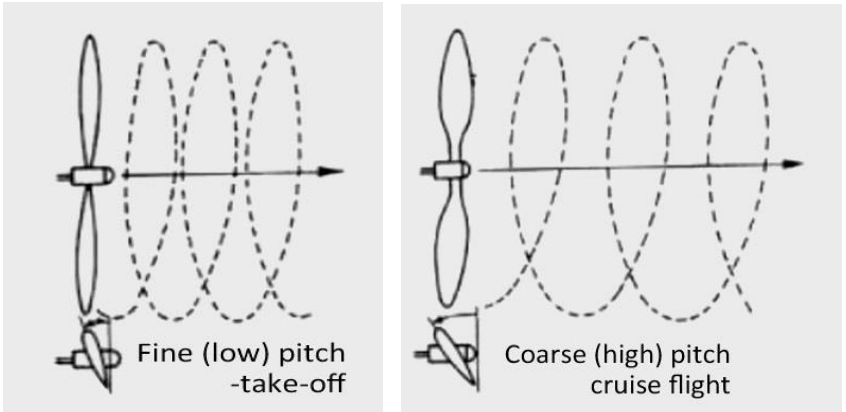


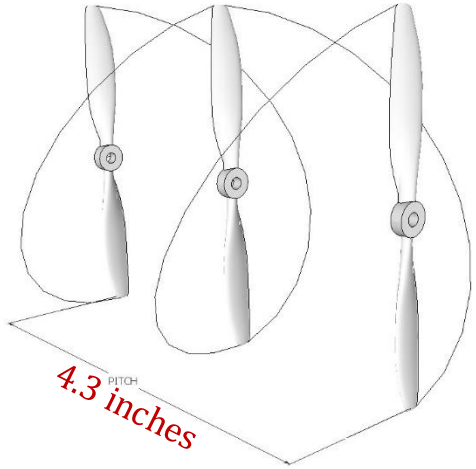
Propeller selection

Diameter: A propeller with a larger diameter enables more contact with the air. As a result, it causes more stability when the drone is hovering. a smaller diameter takes more effort when maneuvering smoothly in the air. Still, they speed up or slow down easily hence being more responsive than their larger counterparts.



Pitch: Traveling distance of the propeller per each revolution. low pitch generates less turbulence and more force in the air. This makes the motors work effortlessly in carrying heavy payloads. Also, it results in increased flight time since the motors use less current from the battery. High pitch produces less force and more turbulence in the air.





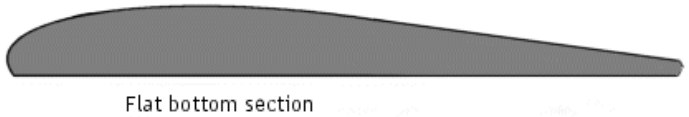
Number of Blades: In terms of speed and efficiency, 2-blade props are better, especially in lightweight drones using less powerful motors. But 3-blade props are a suitable choice when prioritizing stability and extra thrust.

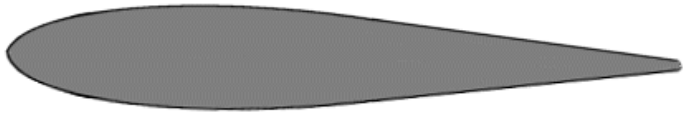


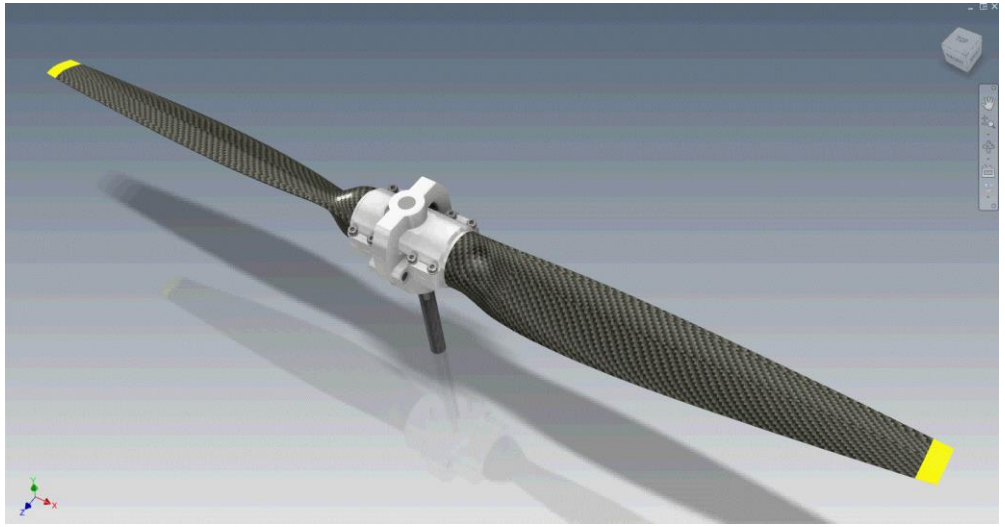
2-Blade Propellers	3-Blade Propellers
More efficient	Powerful with better thrust
Higher speed	Lower speed
Noisy	Less noisy
Reduced stability	Highly stable

Propeller selection

Types of Propellers

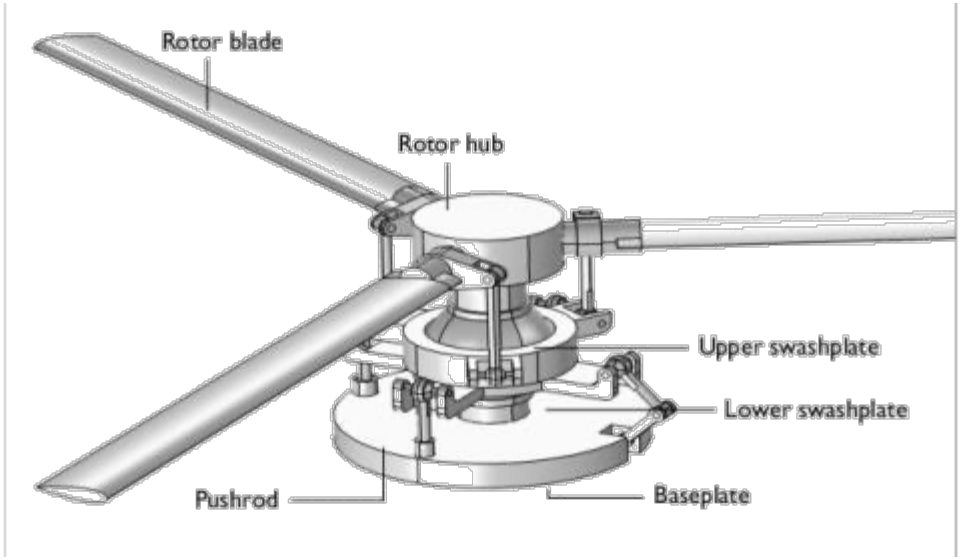






- Multi-rotor drones generally use **non-symmetrical propeller** with fixed pitch
- Helicopter and co-axial helicopter usually use symmetrical propeller with **variable pitch**





Propeller selection

Material of Propellers

- Plastic
- Carbon fiber
- Wood



Factors affecting selection

1. Weight: Light materials reduce drone weight, improving efficiency and increasing payload capacity.
2. Strength/Durability: Materials must withstand forces, impacts, and vibrations without breaking or deforming.
3. Stiffness/Flexibility: Balance is needed for control, stability, and shock absorption during flight.
4. Cost: Striking a balance between performance and affordability is crucial for drone manufacturing.

Propeller installation

Mounting of Propeller on Motor

Direct mounting

- Suitable for all propellers and motors
- Use adapter for different hole size



Propeller saver

- Easy break-off upon collision
- Uses O-ring to secure propeller



Self tighten

- Easy to mount and take off
- Self tighten when rotating



Propeller installation

Propeller Balancing

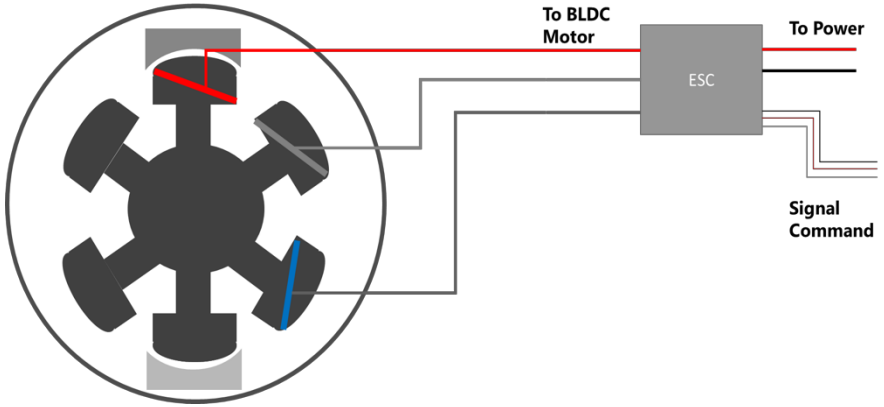
- Propeller needs to be balanced to reduce vibration
- A balancer could be used
- May use sandpaper to sand the surface on heavier side

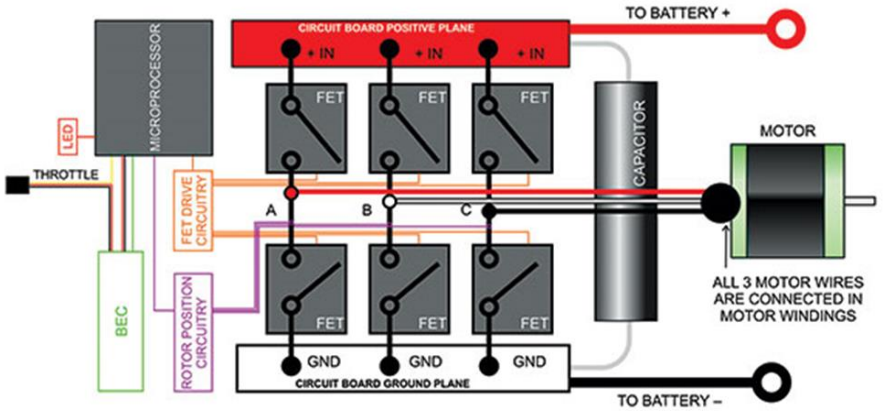


ESC (Electronic Speed Controller) selection

ESC helps to:

- Brake the motor
- Soft start the motor
- Control the speed of the motor





Key Factors:

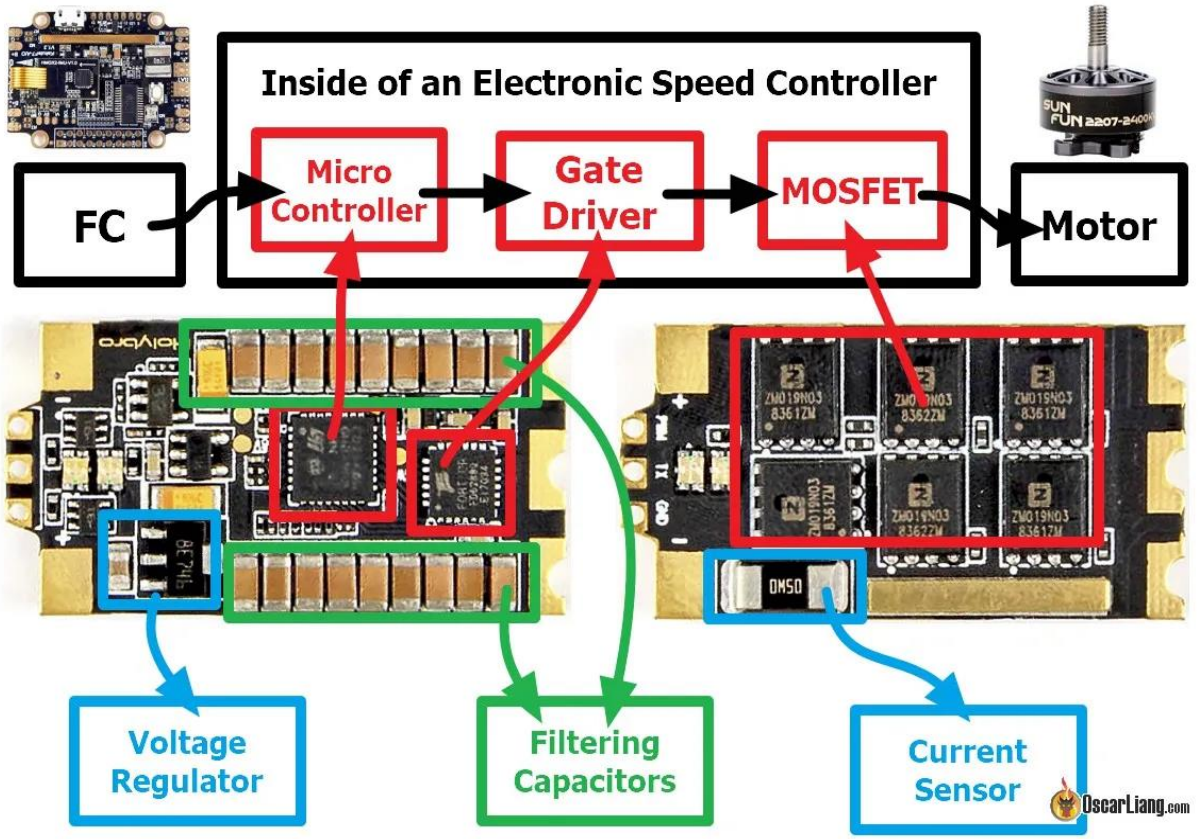
- **Current Rating**: Choose an ESC with a current rating 20-30% higher than the motor's maximum current draw (e.g., if the motor draws 20A, use a 25-30A ESC).
- **Voltage Rating**: Ensure the ESC supports the battery's voltage (e.g., 3S, 4S, etc.).
- **Firmware**: Use ESCs with compatible firmware (e.g., BLHeli or SimonK) for smoother and responsive motor control.
- Type: Opt for high-performance ESCs for racing drones and standard ESCs for general use.



ESC Anatomy

The essential components on an ESC:

- Microcontroller unit (MCU)
- Gate driver
- MOSFET
- Low dropout voltage regulator (LDO)
- Current sensor
- Filtering capacitors



Gate Driver:

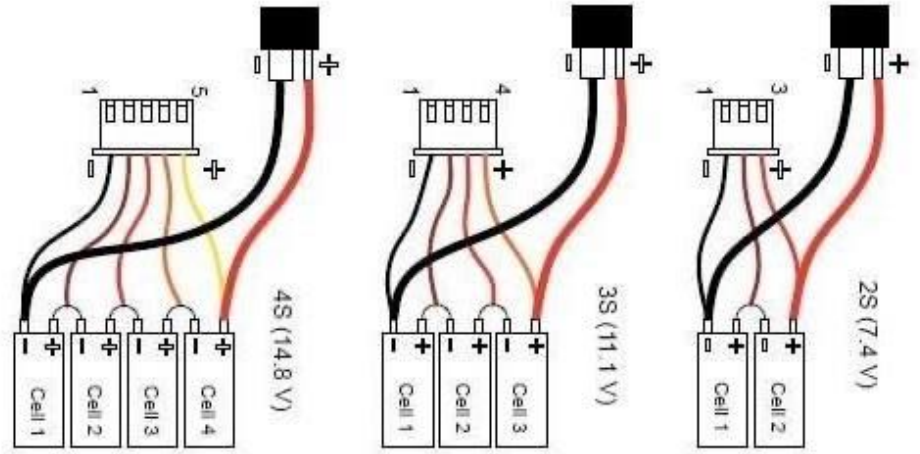
used to amplify low-voltage signals from a microcontroller to high-voltage signals for power switches. This allows the switches to turn on and off quickly, which reduces power loss and increases efficiency.

MOSFET:

MOSFETs are like switches (transistors), switching the power on and off thousands of times per second, to drive motors. Bigger MOSFETs usually mean the ESC can handle higher voltage and current, making the ESC more robust.

Battery selection

- ✓ Commonly nickel based and lithium-based batteries are known
- ✓ Lithium based batteries are mostly used due to best performance.
- ✓ Lithium batteries has two most commonly used batteries are:
  - Lithium-ion battery (e.g., Automobile batteries etc.)
  - **Lithium polymer battery (Li-po) (e.g., Quadcopter , helicopter etc.)**



Using a polymer electrolyte instead of a liquid electrolyte, Li-po batteries are most commonly used in drones due to:

- Less weight
- High energy density
- High discharging capacity
- Can be tailored to the required shape





Attention required for Li-po batteries:

- Heat
- Fully charged for several days
- Over discharging (voltage & current)
- Over charging (voltage & current)
- Improper storage voltage
- Physical damage (dropping, over tightening straps, prying cells apart etc.)

Battery selection

Voltage (S Rating):
Common options are 3S (11.1V), 4S (14.8V), or higher. Higher voltage increases power output but requires compatible motors, ESCs, and propellers.

Capacity (mAh):
Higher capacity batteries offer longer flight times but add more weight. Strike a balance between capacity and total weight.

C-Rating: Determines the discharge rate. A higher C-rating ensures the battery can handle the current demands of the motors and ESCs. It refers to the maximum safe discharge current that a battery can provide (without exploding).

Weight: Ensure the battery's weight doesn't exceed the quadcopter's payload capacity.



$$\text{Required C-Rating} = \frac{\text{Total Current Draw of Motors}}{\text{Battery Capacity in Ah}}$$

Many drones use C-Rating between 15C to 25C.

Battery selection

Battery

Capacity

- Measured in mAh
- e.g.: 1000 mAh meaning if it discharge at 1000 mA, it will last an hour.





Discharge Rate

- Measured in C
- e.g.: 10 C meaning it can discharge at 10 times the capacity of the battery

Battery selection

Battery

If a UAV consume 3.3 A of current, **how long will this battery last**?

- $\frac{2.2}{3.3} = 0.66\ hours = 40\ minutes$



What is the **max output power** for this battery?

- Max Power = max discharge rate × capacity × voltage
- Max Power $= 30 \times 2.2 \times 11.1 = 732.6\ Watt$

Thrust bench (test bench) – benchmarking tool for comprehensive performance analysis

This thrust test aims to measure the thrust and other characteristics of the selected VTOL propulsion system. The test will assess the Motor-Propeller-Battery combination's performance and provide valuable data for analysis and optimization.

Purpose of Test:

- Evaluate the thrust characteristics of the VTOL Motor with various propellers.
- Determine the VTOL combination power output and propeller efficiency at various throttle settings.
- Validate performance specifications and ensure compliance with regulatory requirements.
- Provide data for comparison with manufacturer specifications and performance expectations.

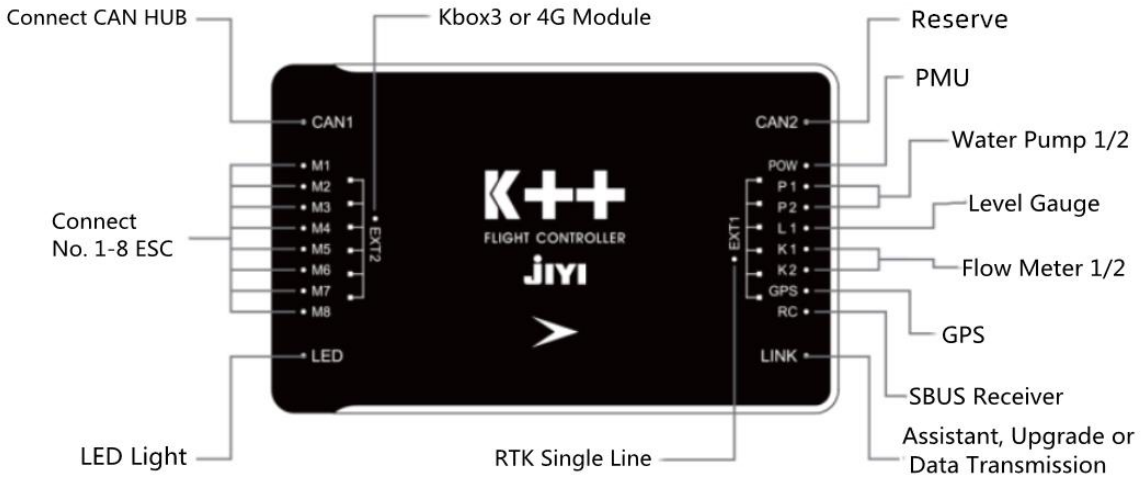


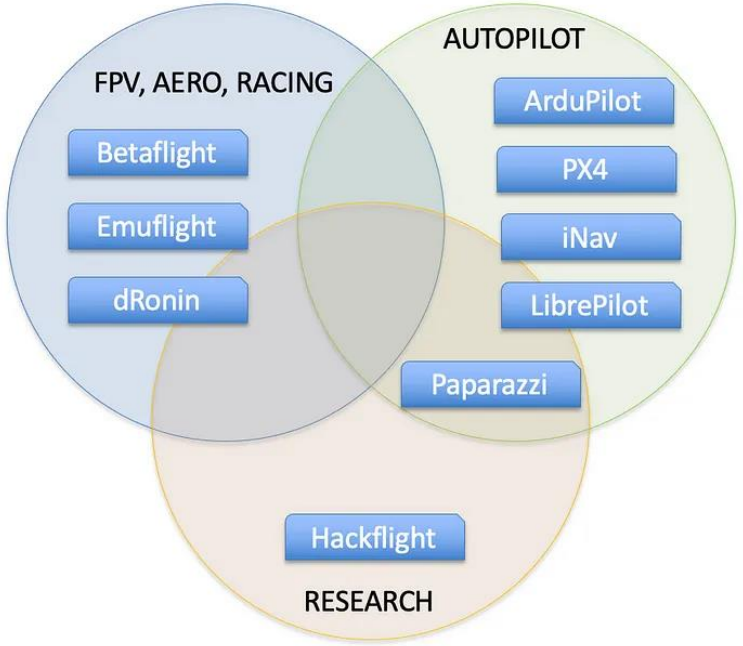
Test Report

Item No.	Prop	Throttle	Voltage (V)	Thrust (g)	Torque (N*m)	Current (A)	RPM	Inputpower (W)	Efficiency (g/W)	Operating Temperature °C)
MN2806-	T-MOTOR P12X4	40%	23.59	230	0.04	0.76	3080	17.85	12.91	45℃
		45%	23.54	282	0.05	1.00	3417	23.62	11.93	
		50%	23.46	344	0.06	1.32	3755	30.94	11.11	
		55%	23.33	398	0.07	1.63	4036	37.94	10.50	
		60%	23.28	458	0.08	2.00	4319	46.61	9.83	
		65%	23.16	511	0.08	2.38	4571	55.18	9.26	
		70%	23.12	568	0.09	2.83	4837	65.34	8.70	
		75%	23.09	642	0.10	3.34	5094	77.08	8.33	
		80%	23.01	705	0.11	3.84	5337	88.41	7.97	
		85%	22.93	774	0.12	4.45	5594	101.97	7.59	
		90%	22.91	845	0.14	5.10	5838	116.77	7.24	
		95%	22.90	915	0.14	5.78	6068	132.32	6.91	
		100%	22.86	958	0.15	6.30	6235	144.03	6.65	

Flight control system

A flight controller is responsible for stabilizing the aircraft, ensuring precise flight manoeuvres, and providing data to the pilot. It's a circuit board equipped with sensors that detect the drone's movements and user commands. With this information, the FC adjusts the speed of the motors to move the drone in the desired direction.





Main Components in FC:

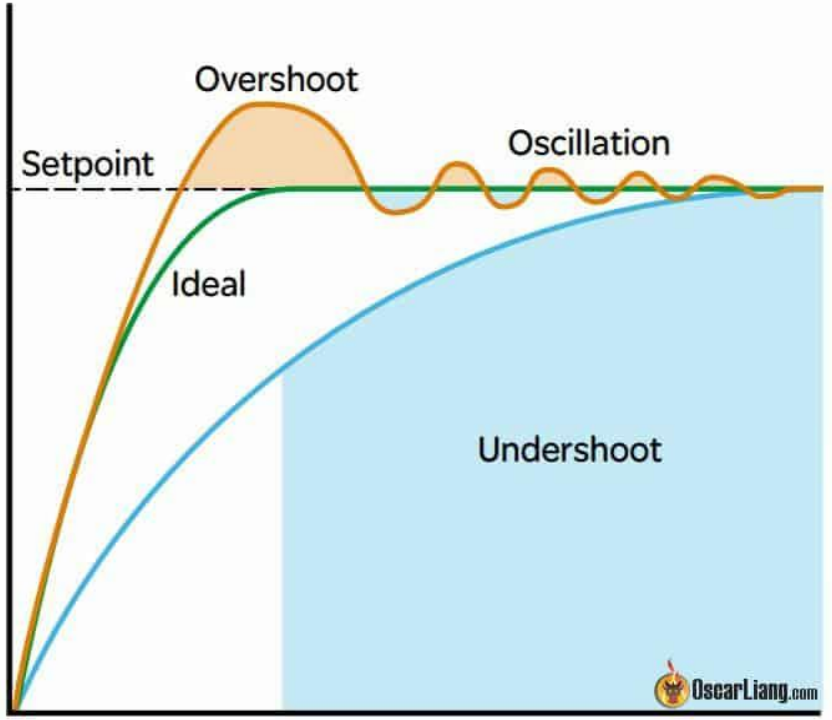
- Microprocessor: Processes inputs from the sensors and the operator
- Internal memory: Stores flight parameters, also known as the Black Box
- Serial interfaces: Allows the flight controller to communicate with other devices (Wi-Fi, Bluetooth etc.)
- Input/Output Interfaces: Allows the flight controller to communicate with the remote control receiver, motors, and other devices (e.g. ESC, LED, SD card slots, USB)
- Sensors: Accelerometer, Gyroscope, Barometer, Compass (Magnetometer)
- Power distribution to each component.

Control method: PID on PX4

PID, which stands for Proportional, Integral, Derivative, is an algorithm within a flight controller's software that reads data from sensors and processes radio stick commands to calculate the required motor speed for achieving the desired rotational rate.

A PID controller's primary goal in a drone is to correct the "error" by adjusting motor speeds. The control loop continuously reads sensor data and calculates motor speeds to minimize the error.

- ✓ P (Proportional) relates to the present error. The larger the error, the harder it pushes – in math term, it's proportional to the error.
- ✓ D (Derivative) predicts future error. It considers how quickly the set-point is approached and counteracts P to minimize overshoot when nearing the target – in math term, it's the derivative of the error.
- ✓ I (Integral) accumulates past errors. It addresses external forces that occur over time, such as a drone drifting away from set-point due to wind or an off-centered weight, by adjusting motor speeds to counteract it – in math term it's the integral of the error.



Flight Control System

Control Signal Flow

Step 1





Manual inputs

or



Autonomous inputs

PPM

Step 2

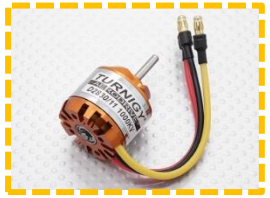


1. Decode input
2. Check measurement
3. Control
4. Calculate output values
5. Send output to ESC

PWM

Step 3



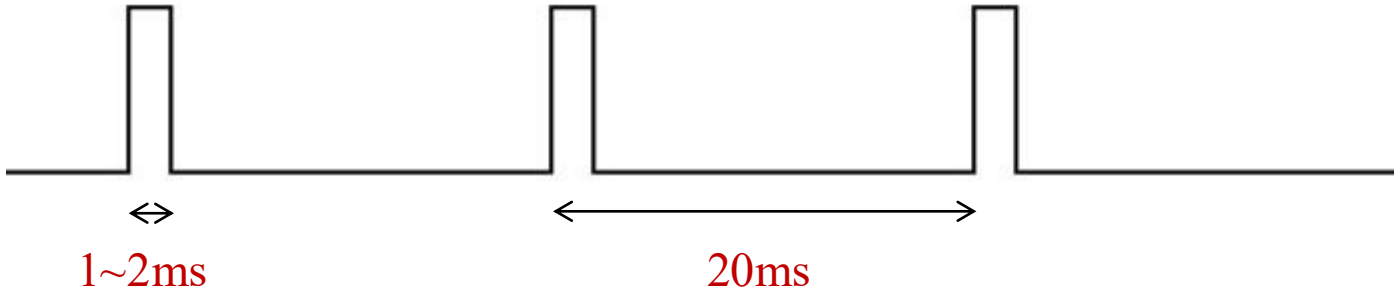


Motors spin with desired speeds

Flight control system

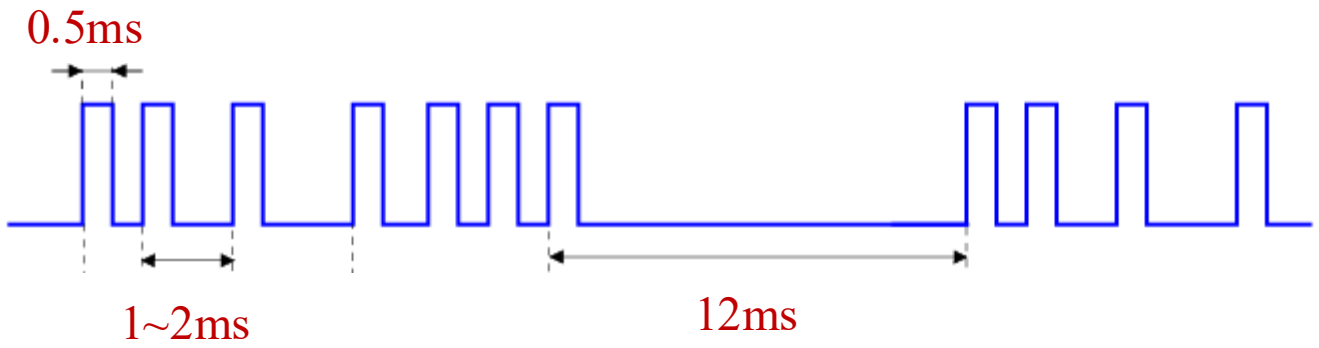
PWM vs PPM

Pulse Width Modulation



- Commonly used in UAV motor control

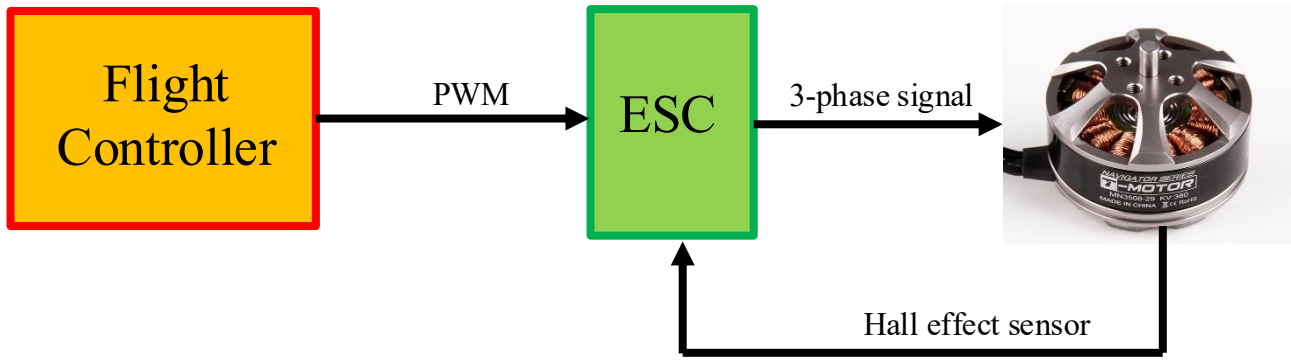
Pulse Position Modulation



- Commonly used in radio communication

UAV Propulsion System

Working Principle



- Flight controller only output PWM signal
- ESC controls the rotating speed of the brushless motor according to PWM signal width
- Hall effect sensor on brushless motor will feedback rotating speed data to ESC for control

IMU Sensors

IMU Sensor: An inertial measurement unit (IMU) is an electronic device that measures and reports a body's specific force, angular rate, and sometimes the orientation of the body, using a combination of accelerometers, gyroscopes, magnetometers:

- **Accelerometer**: It measure the rate of change in an object's velocity, or acceleration. This can include vibrations, bumps, or sharp changes in speed
- **Gyroscope**: a gyroscope can measure the angular rate and angular velocity (accelerometers sense changes in speed and direction. Gyroscopes sense rotational speed)
- **Barometer**: Used to monitor the pressure difference and help in judging the Altitude
- **Compass (Magnetometer)**: Helps in measuring the strength and direction of magnetic fields and it is used for Heading Reference systems.



Global Positional System (GPS) & Other Sensors

- GPS/BeiDou (provides location and time information) to FC for navigation.
- RTK (Real-Time Kinematic): a GPS correction technology.
- Hyperspectral/Multispectral camera
- Ultrasonic sensor
- Thermal camera
- Lidar





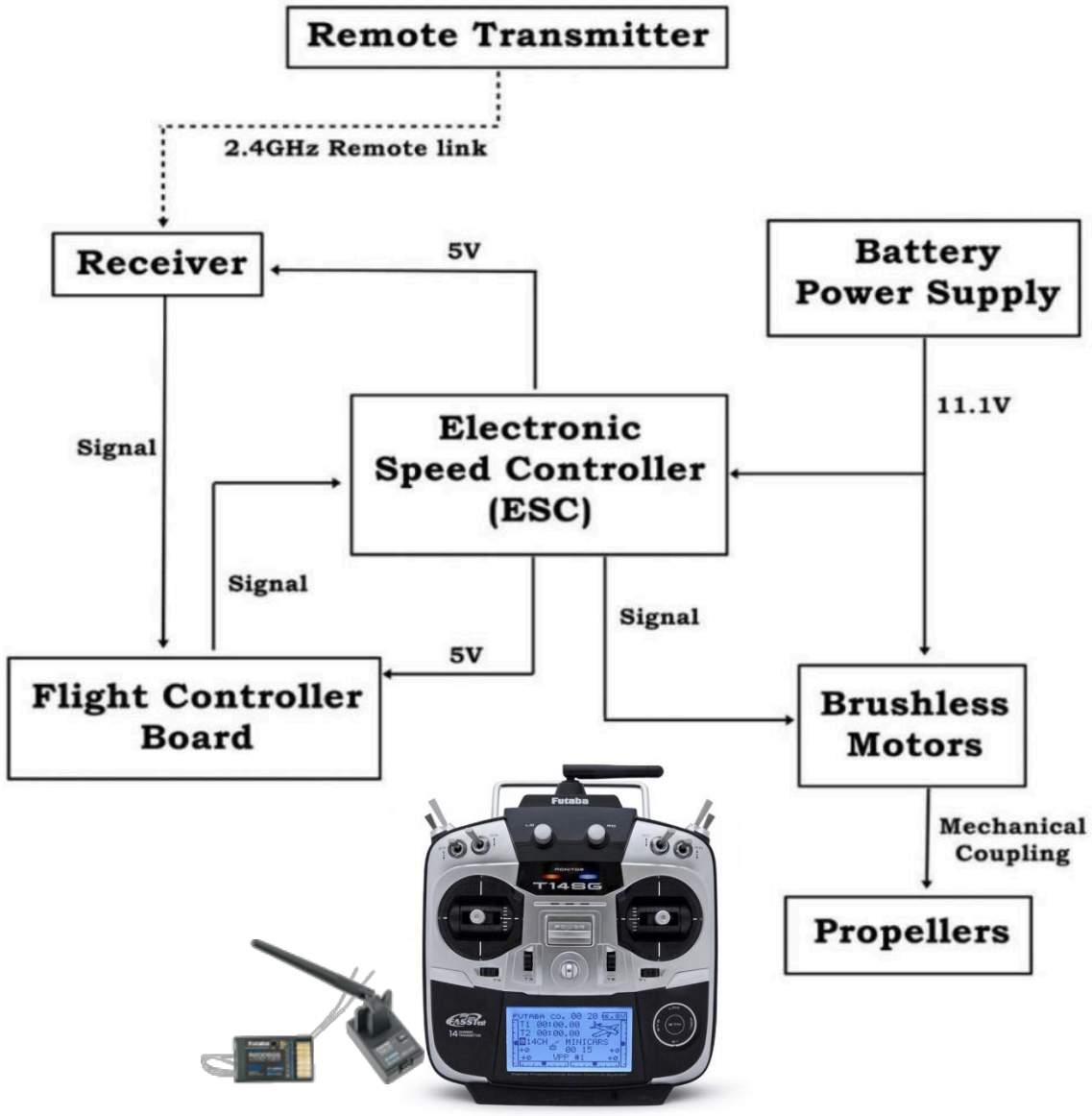








Remote transmitter and receiver



A radio control system is made up of two elements,

- Transmitter – Handheld device
- Receiver - placed on drone.

- The most commonly used drones are 4-channel and 6-channel (Equivalent to number of outputs the transmitter or receiver can give).
- Decode joystick input from pilots, then send the signal to the on-board receiver
- The FLYSKY FS-i6 (X) is mostly available in the market.
- It has a frequency of **2.4GHz** with 2 km distance.
- Different transmitter/receiver might have different way for **binding (transmitter-receiver pair to be linked).**

Frequency

- Usually 72 MHz or 2.4 GHz
- 2.4 GHz advantages: less interference, less power consumption, faster response, higher accuracy
- 2.4 GHz disadvantages: need good line-of-sight, reduced performance if is blocked by big objects



Open-source Transmitter

- Able to customized transmitter function based on preference
- http://www.open-tx.org
- http://www.reseau.org/arduinorc

What is the difference of elevator in VTOL and Fixed-wing?

Four Main Channels

- Aileron – control lateral movement
- **Elevator – control front/back movement**
- Throttle – control flying height
- Rudder – control heading of drone

Power Distribution Board (PDB) and Power Module

- PDB's essentially distribute the power from the battery to other components of the drone.
- It is usually connected to power module which reduces the incoming voltage to 5V.
- It directly gives voltage supply to ESCs, Power Module, Pump Motor etc.





- **Power module** is an electronic component that decreases the voltage to 5V. **BEC in ESC** holds a similar function of power module.
- Once voltage is 5V it is then transferred to Flight Controller
- Also called **PMU** (Power Management Unit)

Telemetry & GCS

- Drone telemetry is data gathered about the aircraft and its surroundings that is sent back to the operator or ground control station (GCS). It is a two-way communication. Common drone radio telemetry frequencies include 433 MHz, 915 MHz and the newer **2.4 GHz.** Wireless telemetry data can also be transmitted via Wi-Fi, which typically has a shorter range than other radio technologies but provides higher data rates.



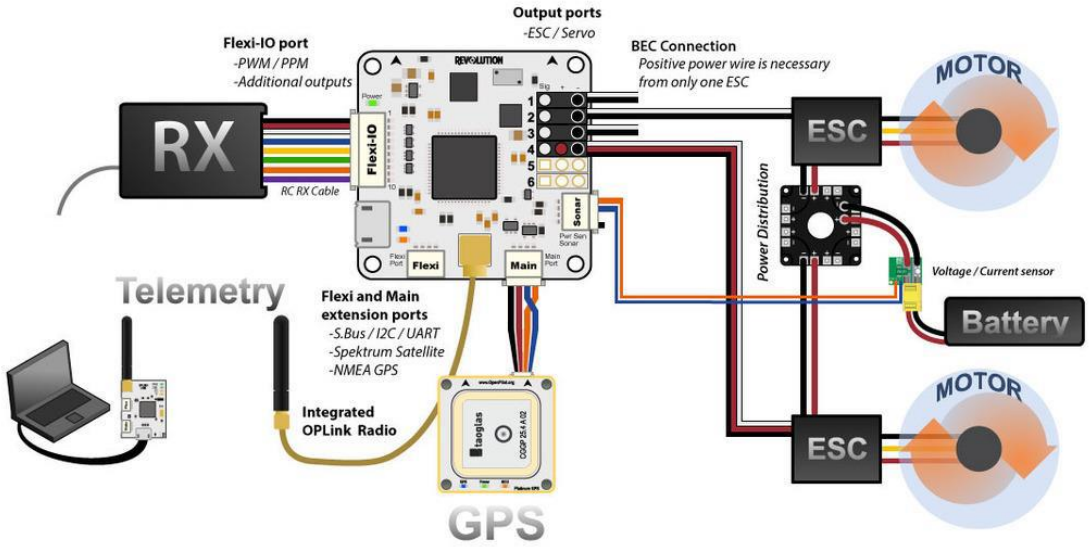
GCS provides the facilities and computers for human control of UAV:

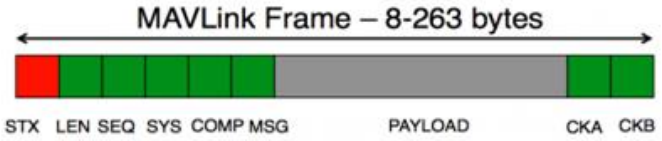
- To operate UAV from ground
- Intuitive inputs with mouse click and key press
- Preset of waypoint for flight, changes of waypoint during flight
- Camera/data display during and after flight



GCS communication with flight control system

- MAVLink Protocol is usually used to communicate between UAVs and GCS.
- MAVLink was developed by Lorenz Meier from ETH based on LGPL (Lesser General Public License) in 2009.
- More information on MAVLink: http://qgroundcontrol.org/mavlink/start





Byte Index	Content	Value	Explanation
0	Packet start sign	v1.0: 0xFE (v0.9: 0x55)	Indicates the start of a new packet.
1	Payload length	0 - 255	Indicates length of the following payload.
2	Packet sequence	0 - 255	Each component counts up his send sequence. Allows to detect packet loss
3	System ID	1 - 255	ID of the SENDING system. Allows to differentiate different MAVs on the same network.
4	Component ID	0 - 255	ID of the SENDING component. Allows to differentiate different components of the same system, e.g. the IMU and the autopilot.
5	Message ID	0 - 255	ID of the message - the id defines what the payload "means" and how it should be correctly decoded.
6 to (n+6)	Data	(0 - 255) bytes	Data of the message, depends on the message id.
(n+7) to (n+8)	Checksum (low byte, high byte)	ITU X.25/SAE AS-4 hash, **excluding packet start sign, so bytes 1..(n+6)** Note: The checksum also includes MAVLINK_CRC_EXTRA (Number computed from message fields. Protects the packet from decoding a different version of the same packet but with different variables).	

Camera Gimbal / Stabilizer

Function

- To ensure camera orientation unaffected by UAV orientation
- Reduce vibration due to UAV movements and rotor disturbance

Working Principle

- 2 to 3 motors/servo motors control orientation of camera
- On-board IMU sensors provide orientation measurement for correction

Important Parameters

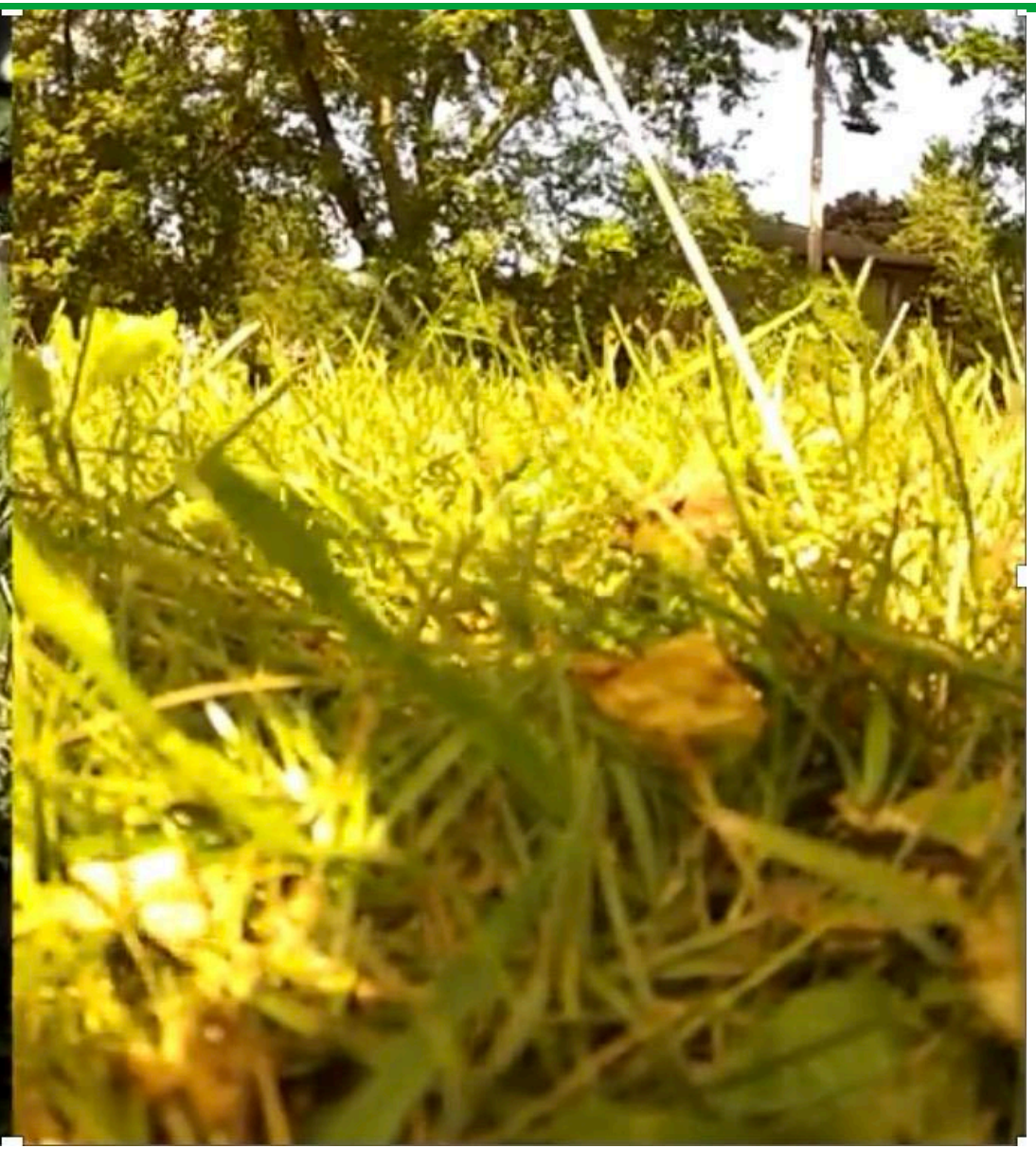
- Weight of the gimbal
- Rated weight of the payload
- Operation Range











UAV Assembly

Peripherals and Wiring

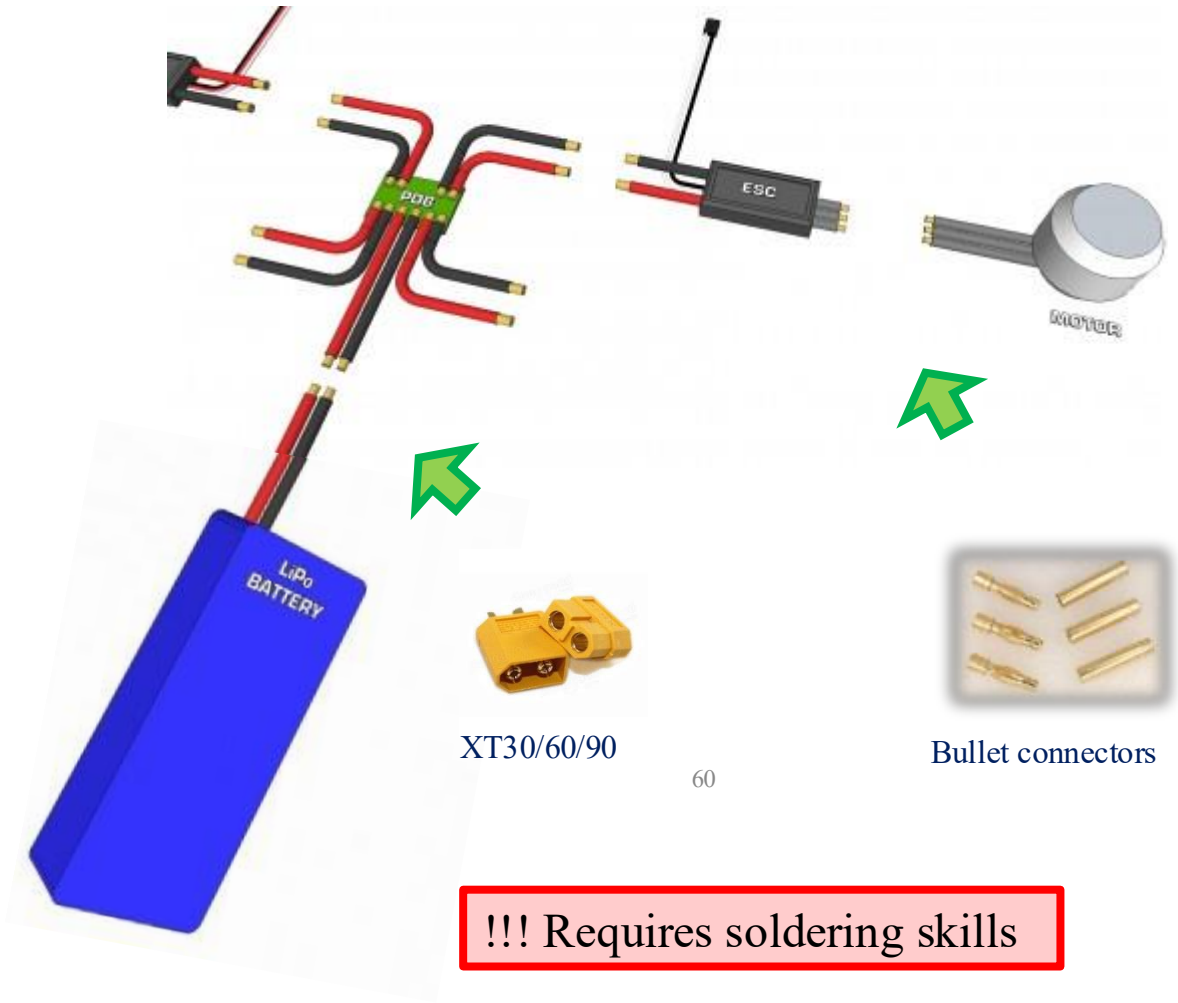


UAV assembly

Step 1

Power line connection

- Battery
- Power distribution board
- ESCs
- Motors



XT30/60/90

Bullet connectors

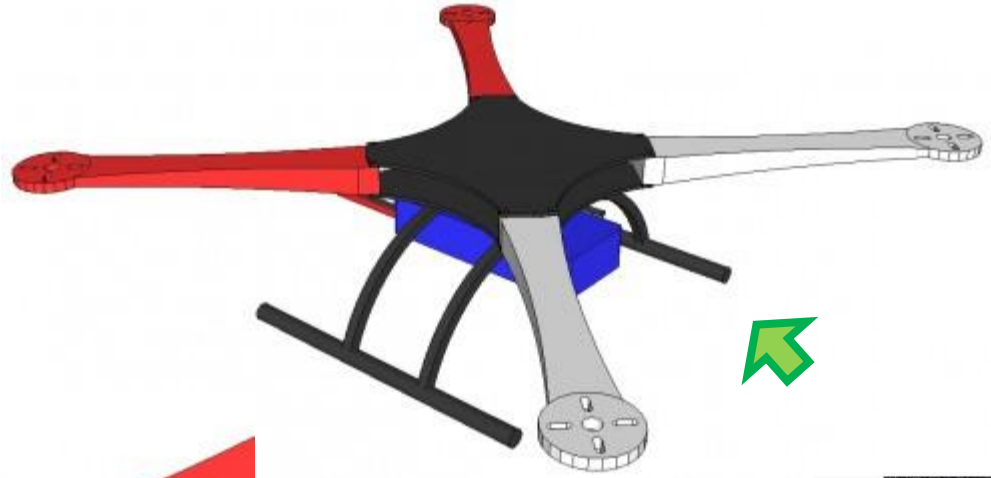
!!! Requires soldering skills

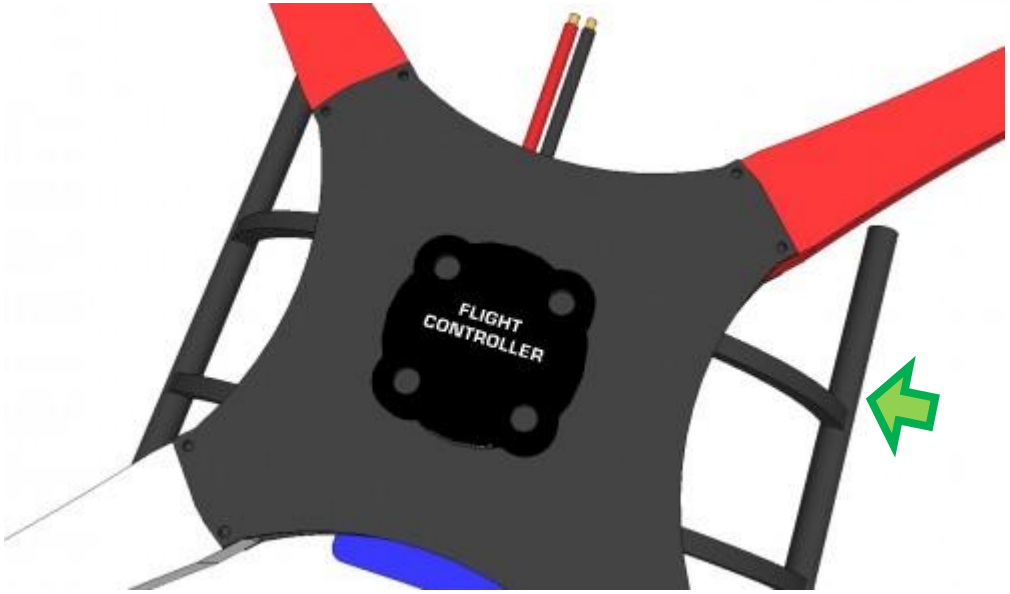
UAV assembly

Step 2

Frame mounting

- Battery
- Flight controller







Velcro



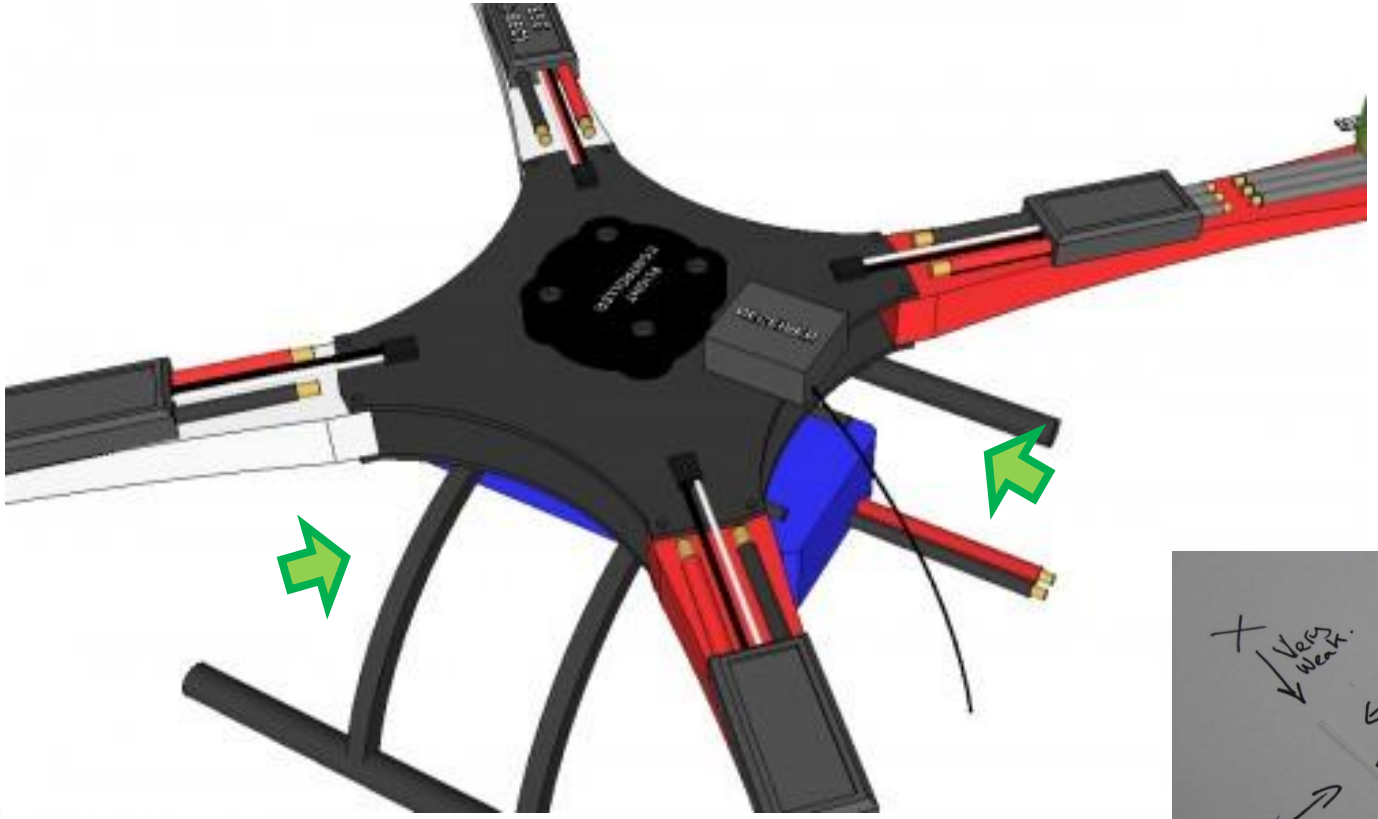
Anti-vibration sponge

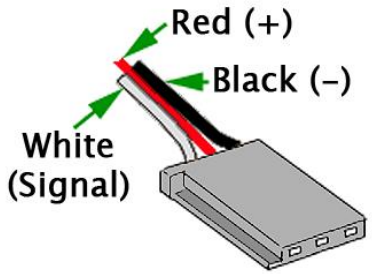
UAV assembly

Step 3

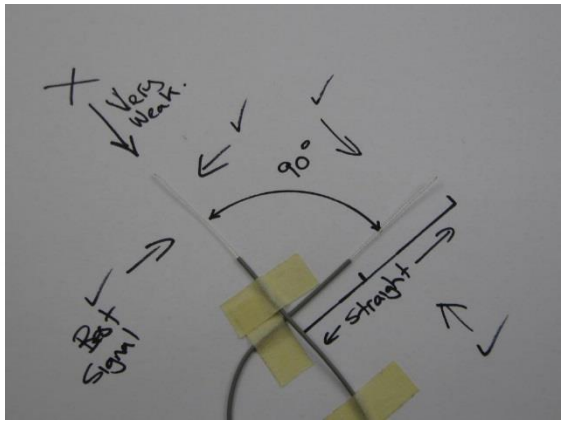
Electronics placement

- ESCs with distribution board
- Receiver
- Connect ESCs to flight controller





Standard servo connector



Optimal 2.4 GHz antenna placement

UAV assembly

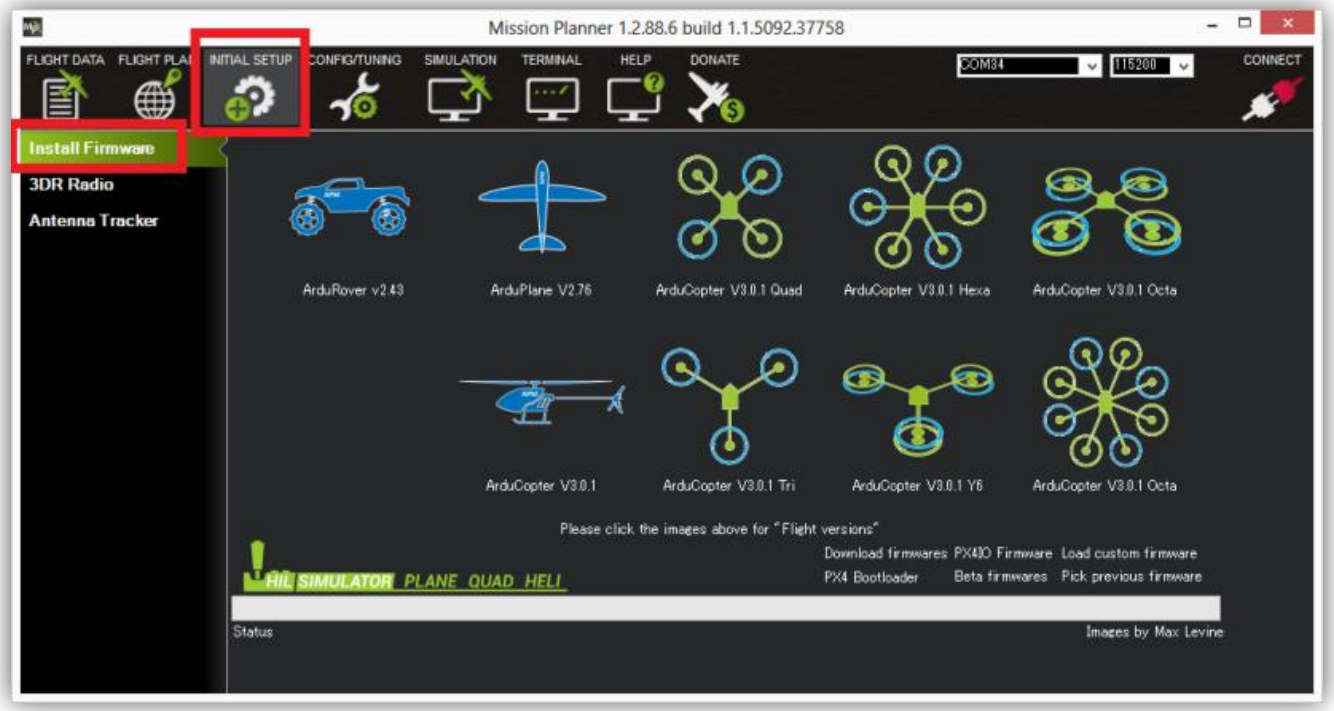
Step 4

Software upload

- Uploading flight control software
- Sensor calibration

Software test

- Runs autopilot program without propeller on
- Test motor's reaction on several case (tilting of drone, yawing of drone …)



RC test

- Test reaction of motors on each different channels input from transmitter
- Range test RC link

Ready to fly (RTF)

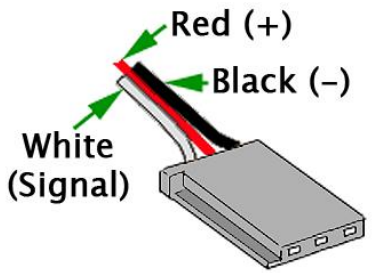
- Put on battery checker (beeper)
- Choose a suitable space
- Fly it!

UAV assembly

Step 5

Connection and Pins





Standard servo connector

BEC (battery eliminator circuit)

- Regulated voltage
- Usually 5 or 6 V

Without BEC

- No regulated voltage


Deans (T-plug)


XT30/60/90

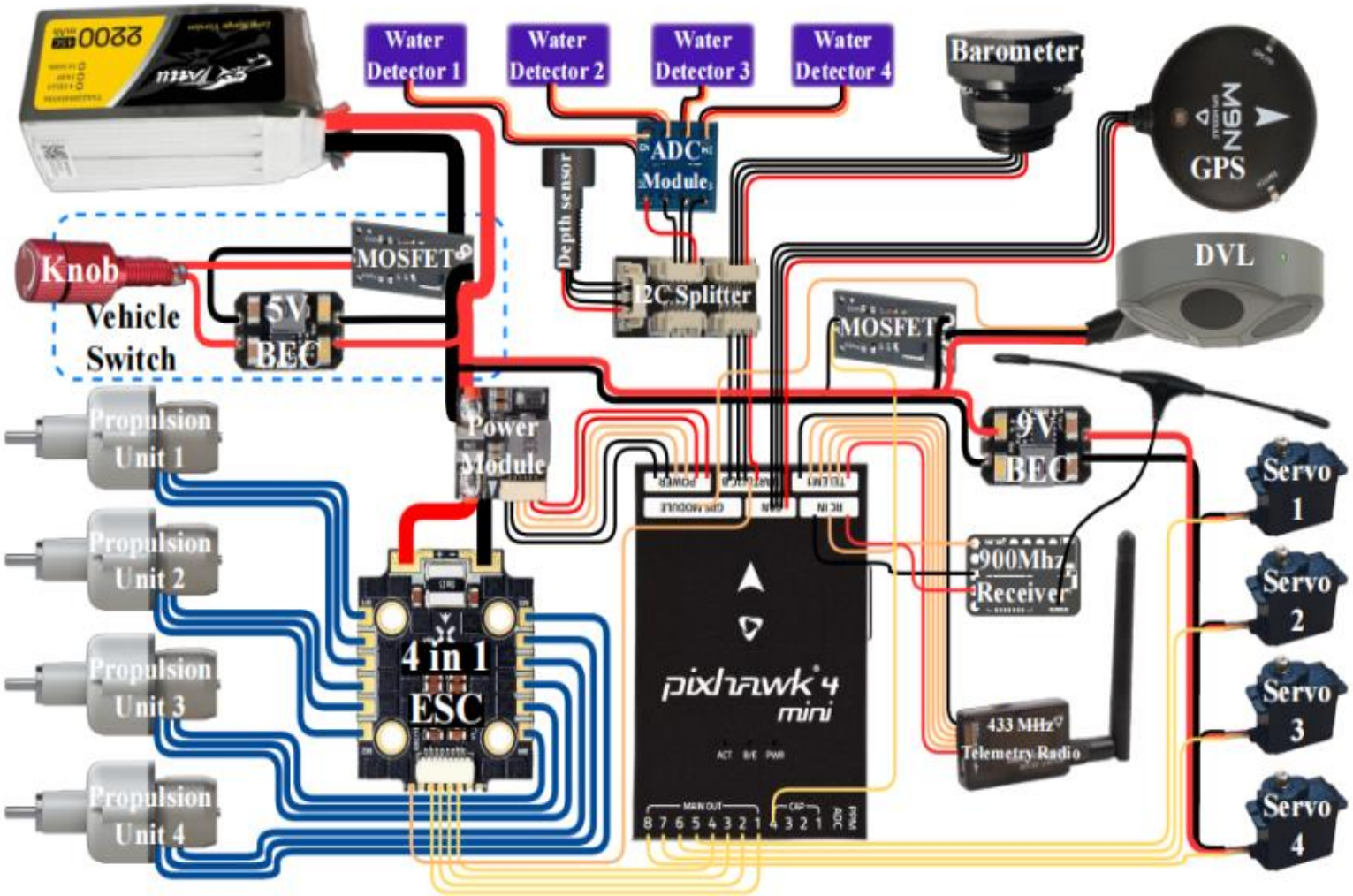

JST


HXT

Example 1: TJ-FlyingFish - Avionic System & Sensors

Using TJ-FlyingFish as an example:

- Barometer for aerial vertical position
- GPS for aerial positioning
- Depth gauge for aquatic vertical position
- DVL for aquatic lateral speed
- More advanced sensors to be integrated…



NUS National University of Singapore

同济大学 TONGJI UNIVERSITY

上海自主智能无人系统科学中心 Shanghai Research Institute for Intelligent Autonomous Systems

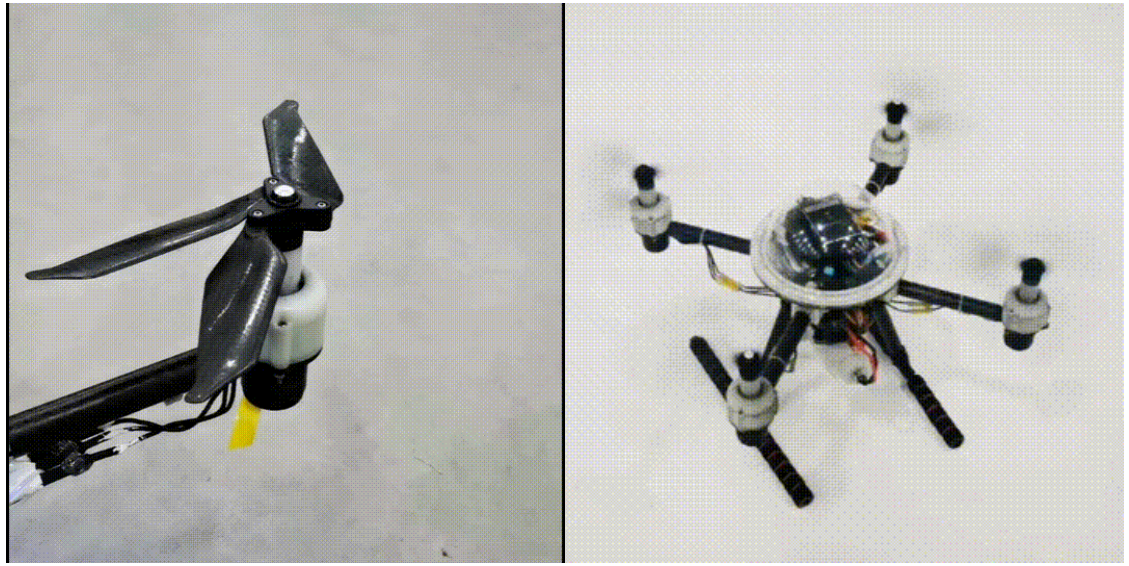
Example: TJ-FlyingFish - Mechanical design

1. An innovative propulsion system is designed for both air and underwater applications

   专为空中和水下应用设计的创新推进系统

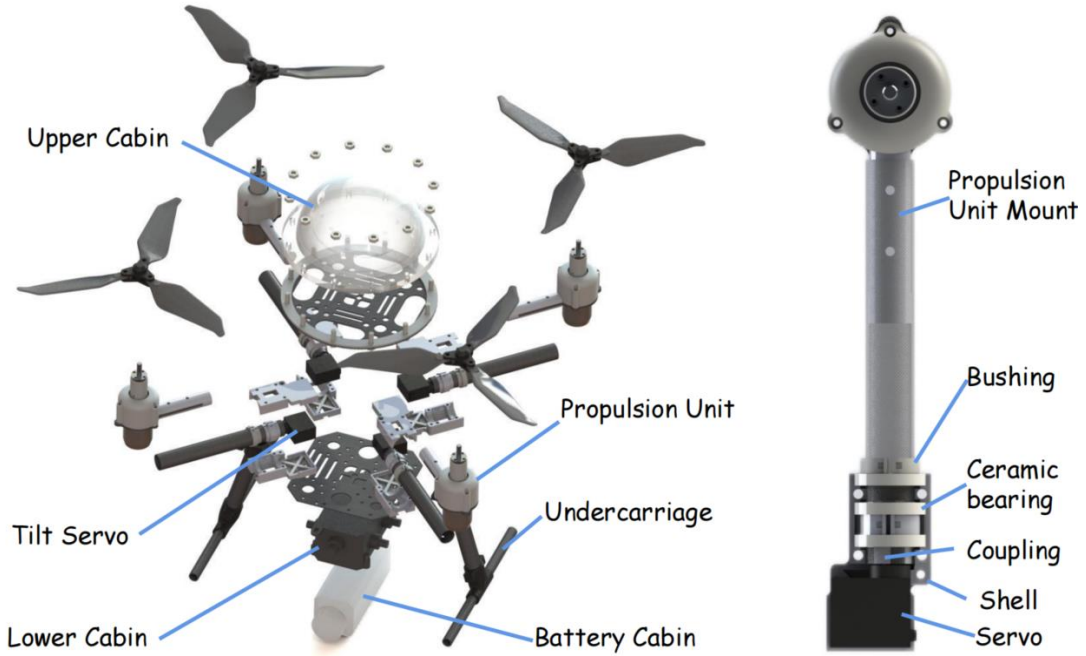
2. Propulsion System: Tilting + aerial-aquatic propulsion unit

   推进系统：倾斜+水空两栖推进装置



Aerial-aquatic (or Cross-medium) propulsion system

水空两栖推进装置



Detailed structure of the platform

平台结构细部

Example: TJ-FlyingFish - Mechanical design

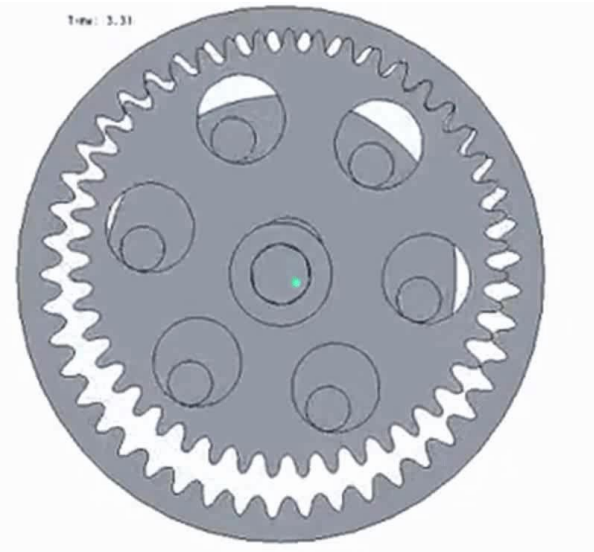
Dual-speed Propulsion Unit 双速推进装置

- Less tooth differential planetary gear transmission

  减齿差行星齿轮传动

- Switching mode by clutch bearing

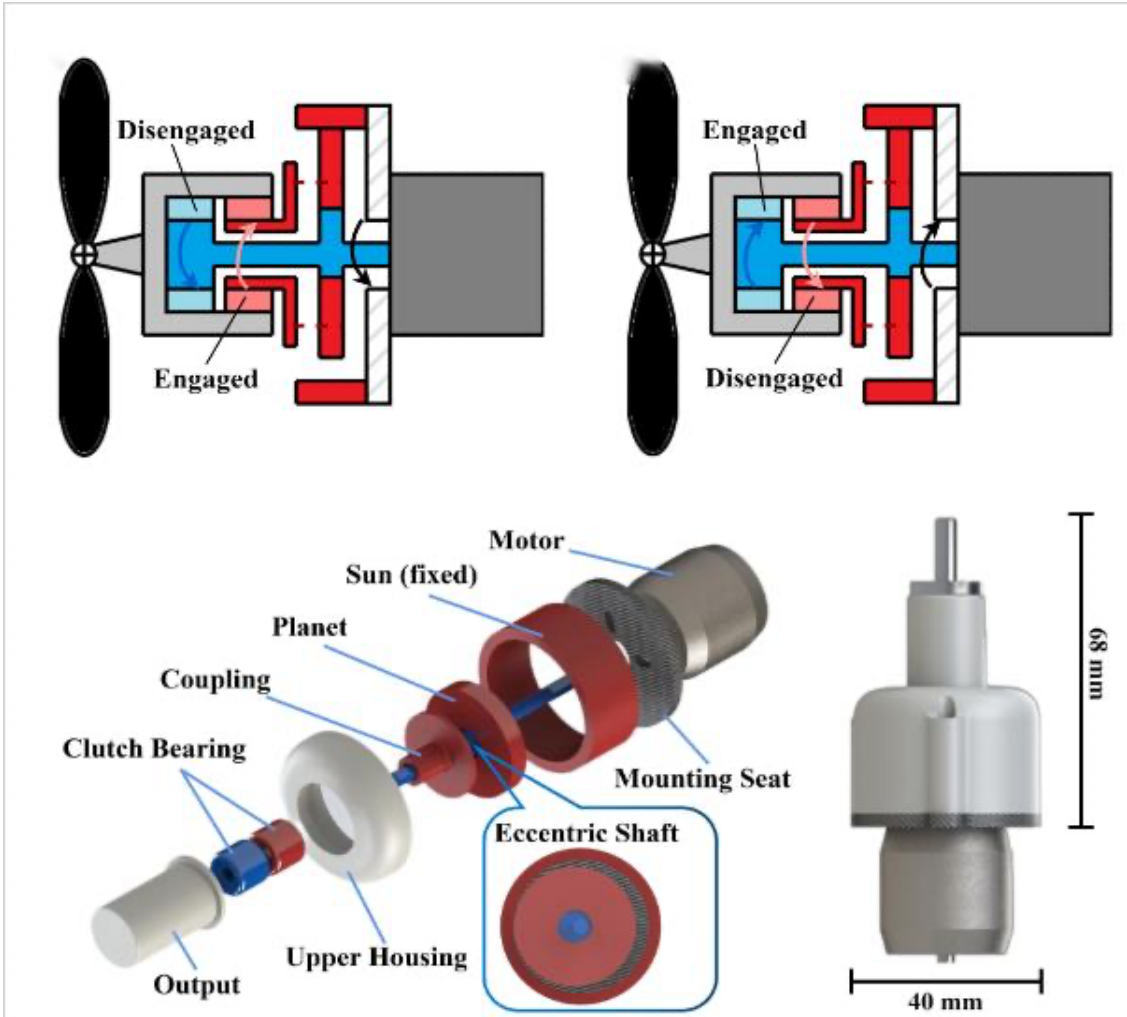
  离合器轴承切换模式





水下模态
Aquatic Mode

空中模态
Aerial Mode



Example: TJ-FlyingFish - Mechanical design

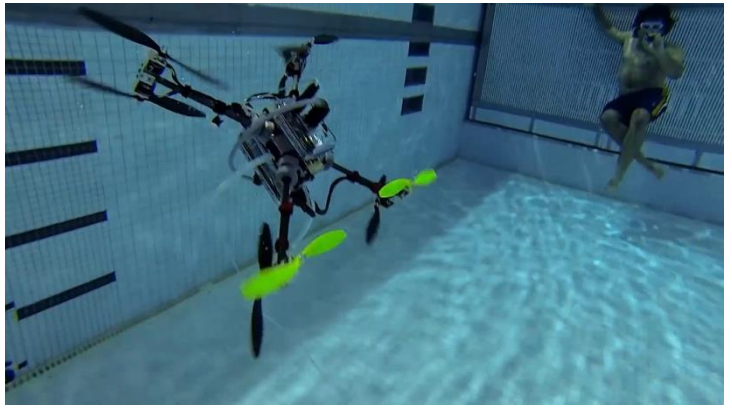
1. Compared to other types of cross-domain vehicles, the motions of our vehicle are much more stable.

   与其他类型的跨域飞行器相比，我们飞行器的运动模态更加稳定

2. This design also brings higher mobility and efficiency 这款设计提升了机动性和效率

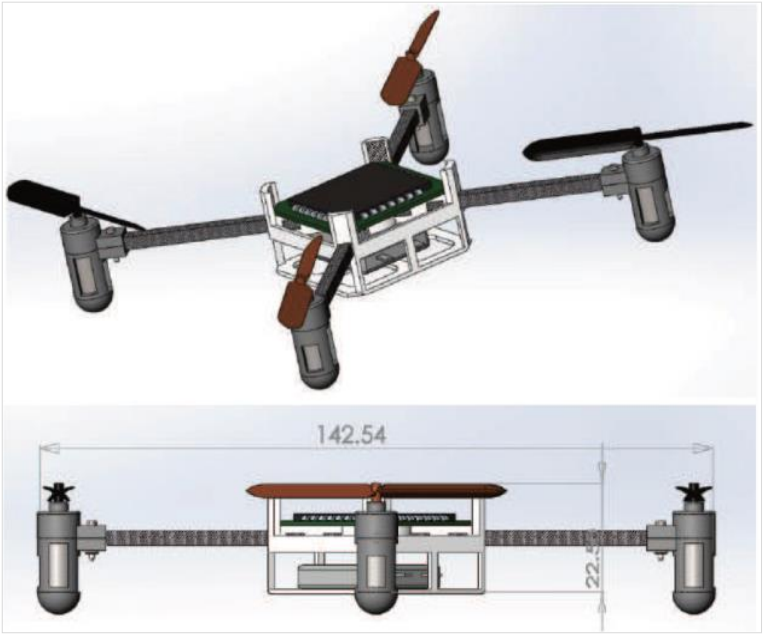


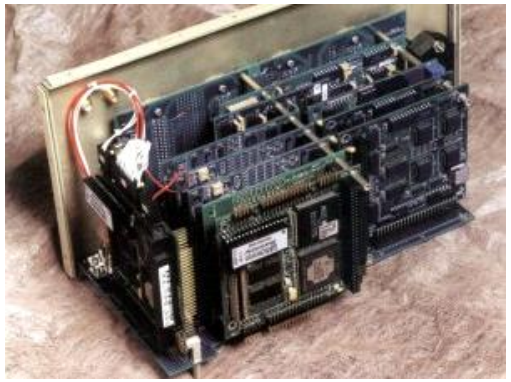


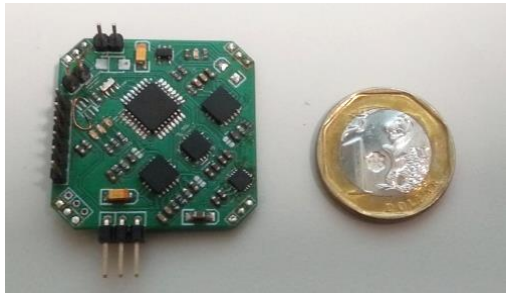


Example: K-Lion - Mechanical design

Cost: SGD 969,996 = HKD 5,819,976



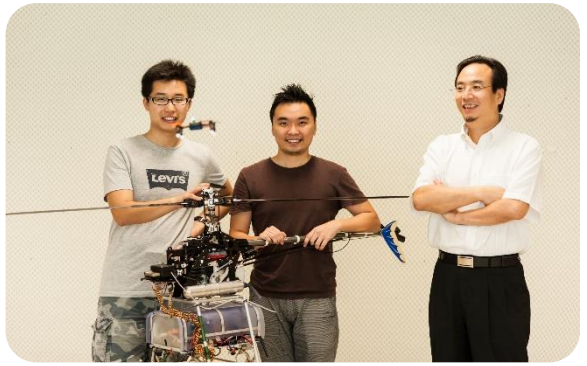




Component	Budgeted Weight (g)	Actual Weight (g)
Power supply:	15	
Battery		9.80
Hardware:	20	
Motor and propeller		14.36
Quadrotor arms		3.72
Quadrotor frame		2.13
Circuit board:	15	
Avionic system		7.32
Camera system		6.25
Miscellaneous:		1.70
Total	50	45.28





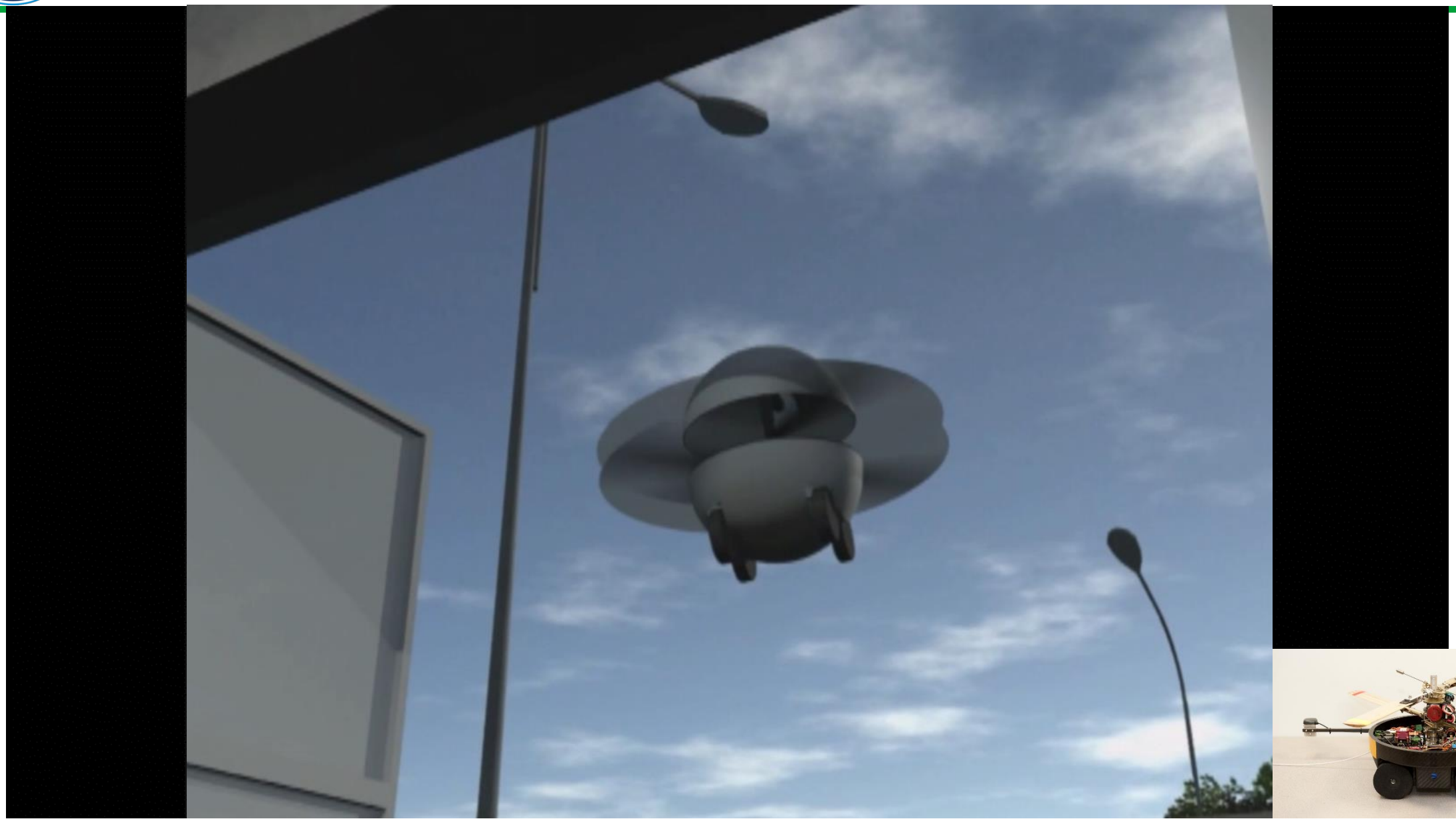


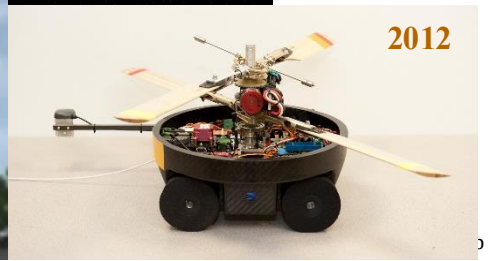
Kun Li, Swee King Phang



Ben M. Chen and his 40-g fully autonomous microaerial vehicle (photograph courtesy of X. Dong).

Example: CoAxial UAV





UAV history

- Gipsy Moth, one of the most common aircrafts in UK in 1929.
- 1935 - Put radio controls on a de Havilland Tiger Moth, a successor to the Gipsy Moth.
- The **Queen Bee (DH.82B)** was one of the first returnable and reusable UAV - used as practice targets.



Aircrafts have commonly been named after insects!



Kettering Bug – 1918
(Pre-dates the name 'Drone')



Radioplane - 1939

UAV history - Queen Bee (DH.82B)



UAV operation – regulations in HK



民航處
CIVIL AVIATION DEPARTMENT

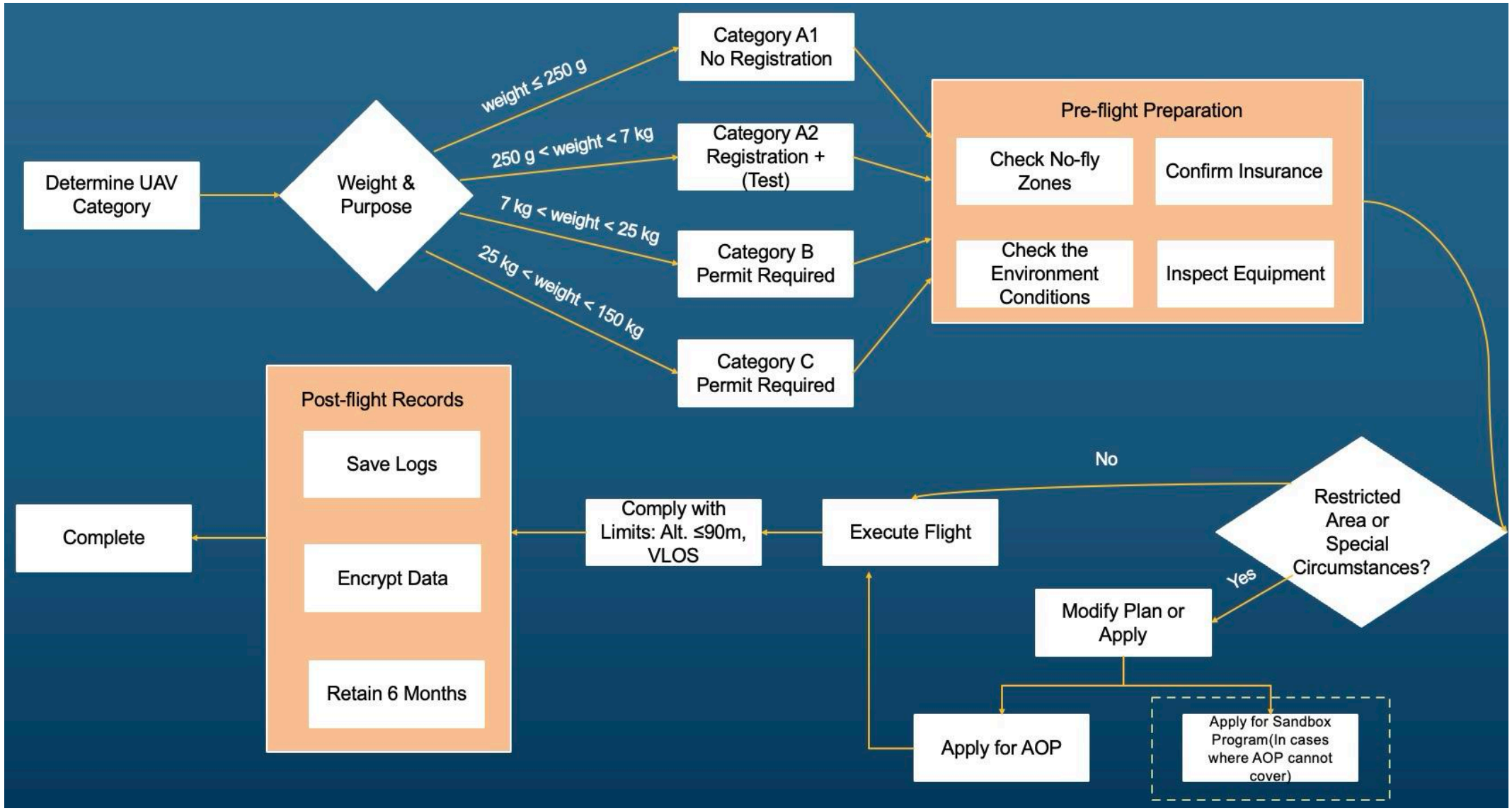


1. Small Unmanned Aircraft Order
2. AOP application process
3. Low-altitude Economy Regulatory Sandbox

UAV operation – regulations in HK – workflow

1. Drone registered?

2. Pilot registered?



UAV operation – regulations in HK - safety requirements

To flight a drone legally in Hong Kong. You have to browse through CAD's website and complete registration of your drone and yourself as a remote pilot:
https://esua.cad.gov.hk/web/





We must comply with the revised **Safety Requirements Document** for Small Unmanned Aircraft Operations issued by the Hong Kong Civil Aviation Department in July 2025.
https://www.cad.gov.hk/documents/Safety_Requirements_Document.pdf

Upon successful registration, a label will be issued, which should be affixed to the aircraft!

Category of Operations	Standard Operations		Advanced Operations
Criteria	Cat A1 SUA (weight≤ 250 g) *and* (Within Cat A1 SUA operating requirements)	Cat A2 SUA (250 g < weight ≤ 7 kg)* *and* (Within Cat A2 SUA operating requirements)	(i) Cat A1 or A2 SUA but exceeding the applicable operating requirements^; (ii) Cross-boundary operation; (iii) Cat B SUA (7 kg < weight ≤ 25 kg); (iv) Cat C SUA (25 kg < weight ≤ 150 kg); (v) Operations involving carriage of dangerous goods; *or* (vi) Operations in restricted flying zone^ ^except wholly within enclosed area
Registration and Labelling Requirements (see Chapter 3)			
Registration & labelling of SUA	×	✓	✓
Min. age of SUA responsible person	×	18	18
Registration of Remote Pilots (see Chapter 4)			
Registration of remote pilot	×	✓	✓
Min. age of remote pilot	×	14	Cat A/B SUA : 14 Cat C SUA : 18
Advanced Rating Training and Assessment Requirements (see Chapter 5)			
Training & assessment of remote pilots with Advanced Rating	×	×	✓
Equipment Requirements (see Chapter 6)			
Safety System (flight log & geo-awareness)	×	✓ (6 months)	✓ (6 months & accessible within HK)
Operating Requirements (see Chapter 7 and Chapter 8)			
Operating requirements (including to maintain the flight wholly within Hong Kong)	✓ (Cat A1 SUA operating requirements)	✓ (Cat A2 SUA operating requirements)	✓ (As specified in Permission)
Permission from the CAD prior to operations	×	×	✓
Insurance Requirements (see Chapter 9)			
SUA insurance for third-party liability (bodily injury and/or death)	×	To be applicable at a later time (1)	✓
Min. coverage	×	HKD 5 million (1)	Cat A/B SUA: HKD 10 million

Table 3 – An Overview of Operating Requirements for Categories A1 SUA and A2 SUA

Operating Requirements	For Category A1 SUA (weight ≤ 250 g)	For Category A2 SUA* (250 g < weight ≤ 7 kg)	
Time of operations	Daylight hours only		
Maintain visual line of sight (VLOS)	✓		
Maximum flying altitude Above Ground Level (AGL)	100 ft	300 ft	
Maximum speed	20 km/hr	20 km/hr	50 km/hr
Minimum lateral separation from uninvolved people / structures / vehicles / vessels	10 m	10 m	30 m
Maximum number of SUA to be operated at the same time by the same remote pilot	1		
Maximum dimensions of SUA	1 m, except that the longest distance between any two rotor blade tips can be up to 1.2 m.		
Carriage of person or animal	No		
Dropping from SUA	None		
Operation Area	Wholly within Hong Kong		

For the avoidance of doubt, unless otherwise specified or permitted by the CAD, the operating requirements applicable to **Category B SUA are the same as those involving Category A2 SUA**. For the operating requirements applicable to Category C SUA, please refer to the AC "**Permission for Category C** Small Unmanned Aircraft Operations and Predefined Scenario #1" (No. **AC-014**) published on the CAD website. Advanced operations require prior permission from the CAD.
https://www.cad.gov.hk/documents/AC-014.pdf



In addition to the provisions in the safety requirements document, the eight points mentioned in the general safety guidelines for Flying SUA must also be observed.
For detailed instructions, please refer to the official website.

https://www.cad.gov.hk/english/sua_safetyreg_safetyguidelines.html

01 Where you cannot fly

02 Choice of Flying Sites

03 Time of Operations

04 Altitude

05 Weather Criteria

06 Heavy SUA

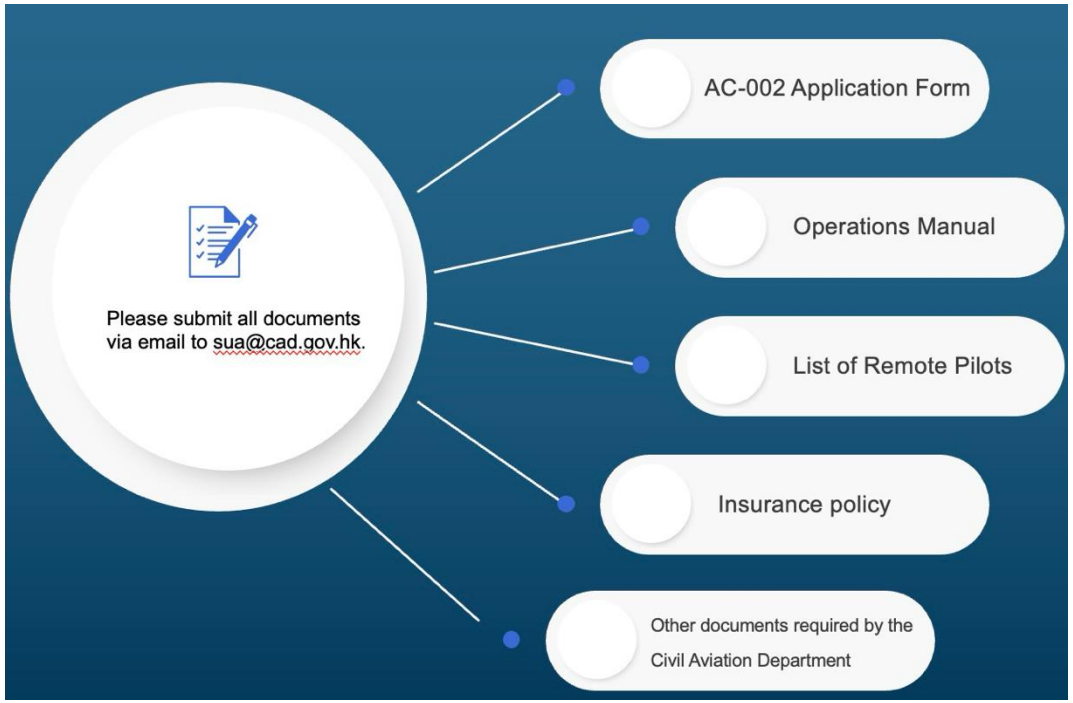
07 Control of SUA

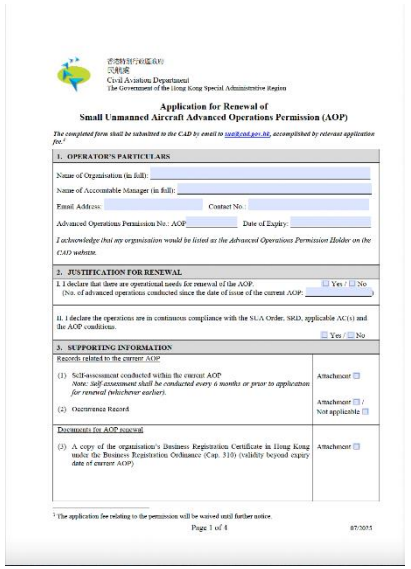
08 Pre-flight preparation

UAV operation – regulations in HK - AOP

If any operation exceeds the limits specified in the regulations, an application for **Advanced Operations Permission (AOP)** must be submitted to the Civil Aviation Department in advance. For detailed instructions, please refer to the official website: https://www.cad.gov.hk/documents/**AC-002**_E.pdf

The application form can be downloaded and completed via the link below.
https://www.cad.gov.hk/application/AC-002_AOP_Variation_Application_Form.pdf




Civil Aviation Department
The Government of the Hong Kong Special Administrative Region

Application for Renewal of Small Unmanned Aircraft Advanced Operations Permission (AOP)

The completed form shall be submitted to the CAD by email to sua@cad.gov.hk, accomplished by relevant application fee.[1]

1. OPERATOR'S PARTICULARS

Name of Organisation (in full):
Name of Accountable Manager (in full):
Email Address: Contact No.:
Advanced Operations Permission No.: AOP Date of Expiry:
I acknowledge that my organisation would be listed as the Advanced Operations Permission Holder on the CAD website.

2. JUSTIFICATION FOR RENEWAL

I. I declare that there are operational needs for renewal of the AOP ☐ Yes / ☐ No
(No. of advanced operations conducted since the date of issue of the current AOP:)

II. I declare the operations are in continuous compliance with the SUA Order, SRD, applicable AC(s) and the AOP conditions. ☐ Yes / ☐ No

3. SUPPORTING INFORMATION

Records related to the current AOP

(1) Self-assessment conducted within the current AOP — Attachment ☐
Note: Self-assessment shall be conducted every 6 months or prior to application for renewal (whichever earlier).

(2) Occurrence Record — Attachment ☐ / Not applicable ☐

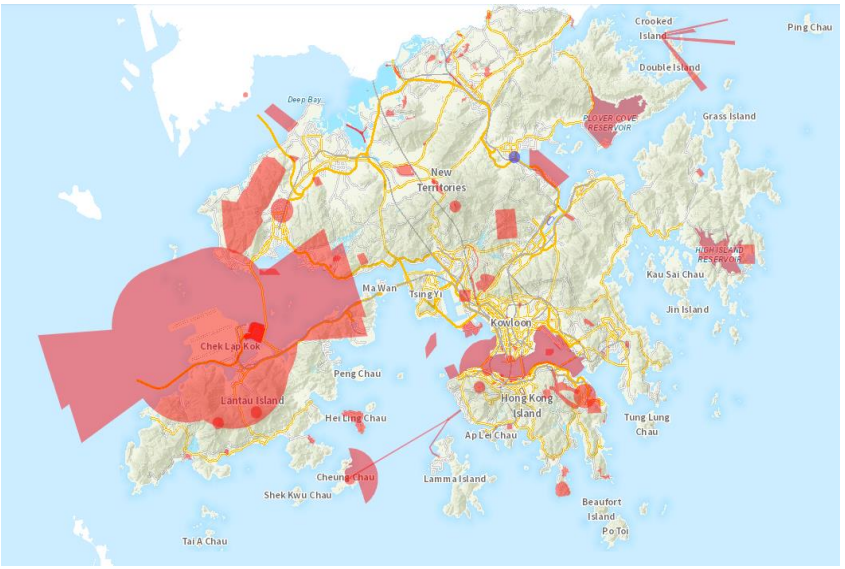
Documents for AOP renewal

(3) A copy of the organisation's Business Registration Certificate in Hong Kong under the Business Registration Ordinance (Cap. 310) (validity beyond expiry date of current AOP) — Attachment ☐

[1] The application fee relating to the permission will be waived until further notice.

Page 1 of 4 07/2025

AC-002



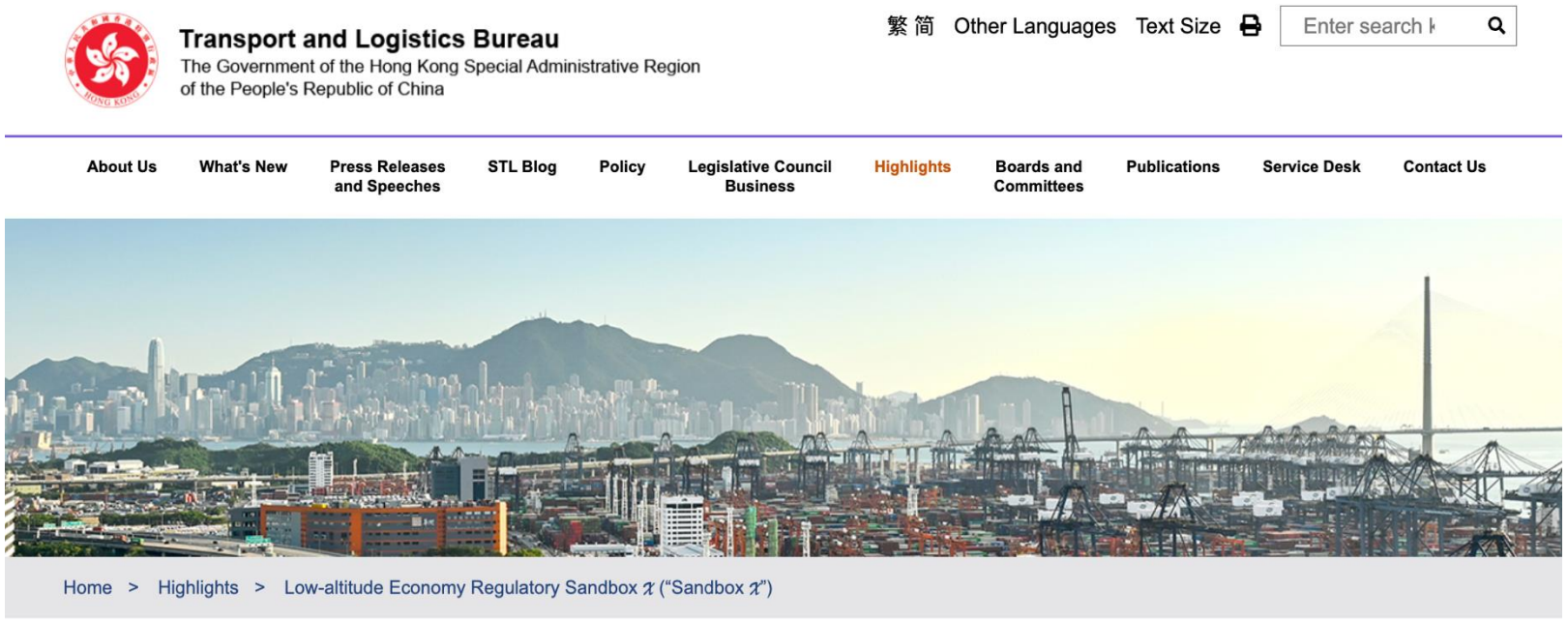
RFZ map

UAV operation – regulations in HK – LAE Sandbox

The purpose of the regulatory sandbox is to allow industry stakeholders to conduct testing and trials of project concepts in a controlled and safe manner within designated routes.

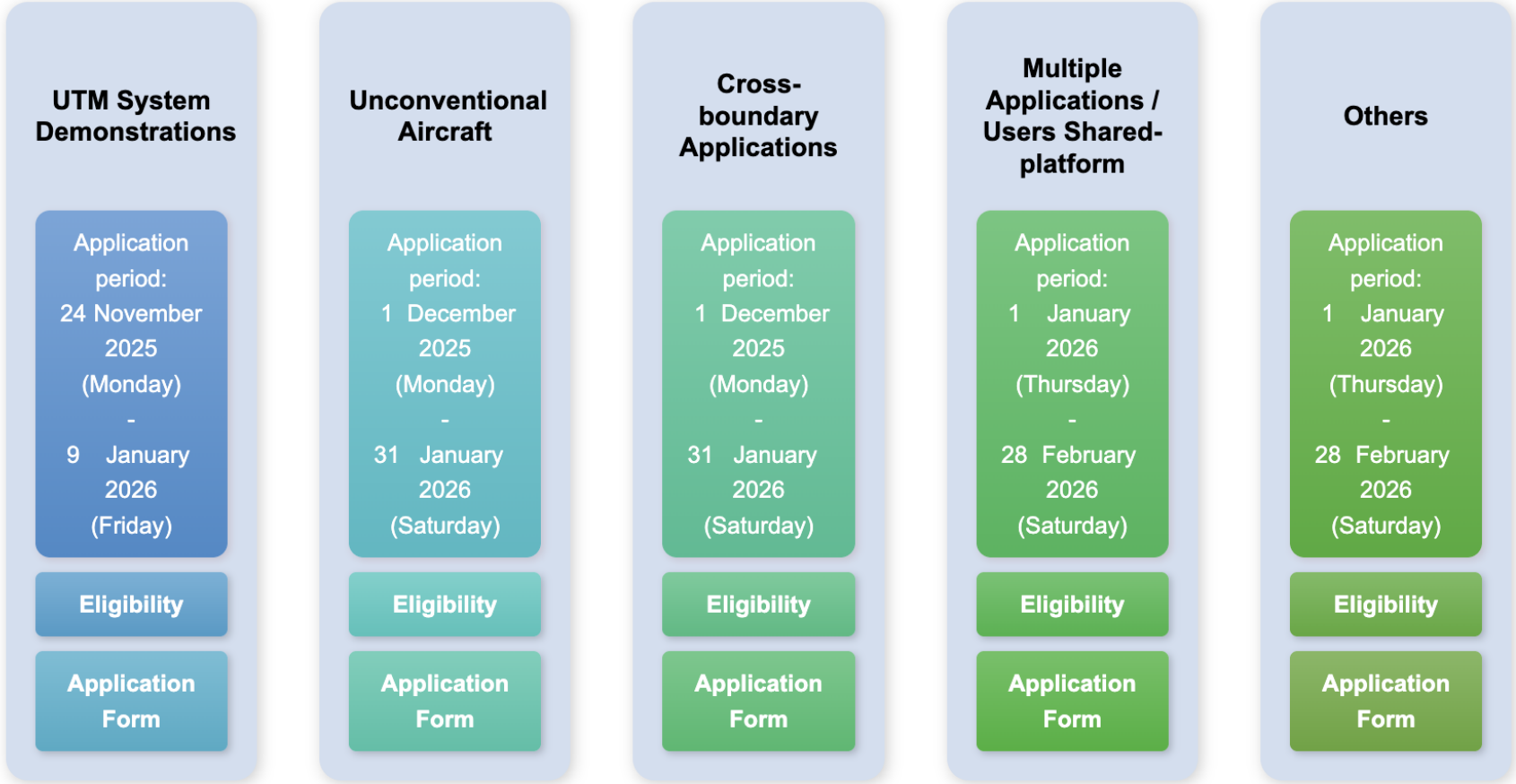
If the company's business involves cutting-edge technologies, such as beyond visual line of sight (BVLOS) operations, autonomous aircraft, or cross-domain integration, which exceed the scope covered by the AOP, it is necessary to apply for the government's regulatory sandbox as a supplementary pathway.

https://www.tlb.gov.hk/tc/highlights/transport/low-altitude.html



The Sandbox χ will be open for applications from eligible parties by modules according to the following schedule:

The regulatory sandbox is divided into the five categories shown in the image on the right. If you intend to apply, you should first check whether you meet the eligibility criteria for the relevant category, then complete the corresponding application form and submit it via email to: **LAE.sandbox@tlb.gov.hk**



UAS operation key terms

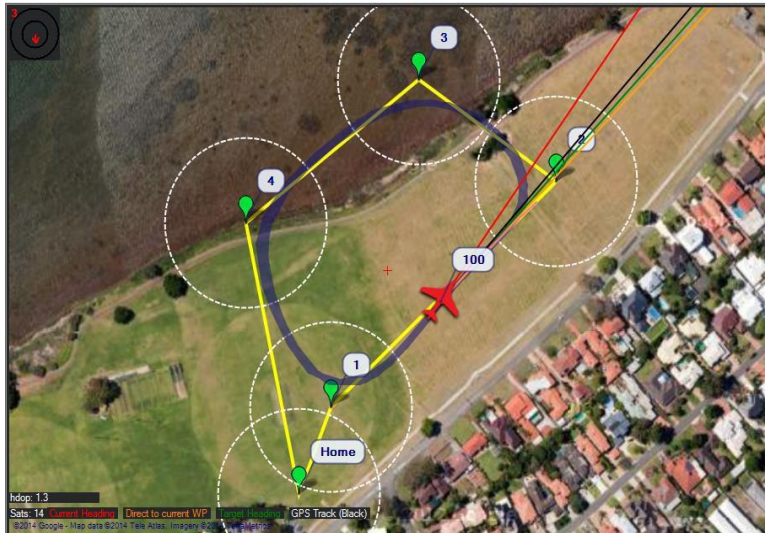
Visual Observer (VO)

- Secondary person to watch over the drone and any potential obstacles around the drone
- Common practice to have a VO for AOP



Waypoint Navigation:

UAV flies as per planned GPS points



UAS operation key terms

Manual Control:

- Flies not by Waypoint Navigation
- Someone is controlling the motion directly



Payload: Equipment being carried, such as:

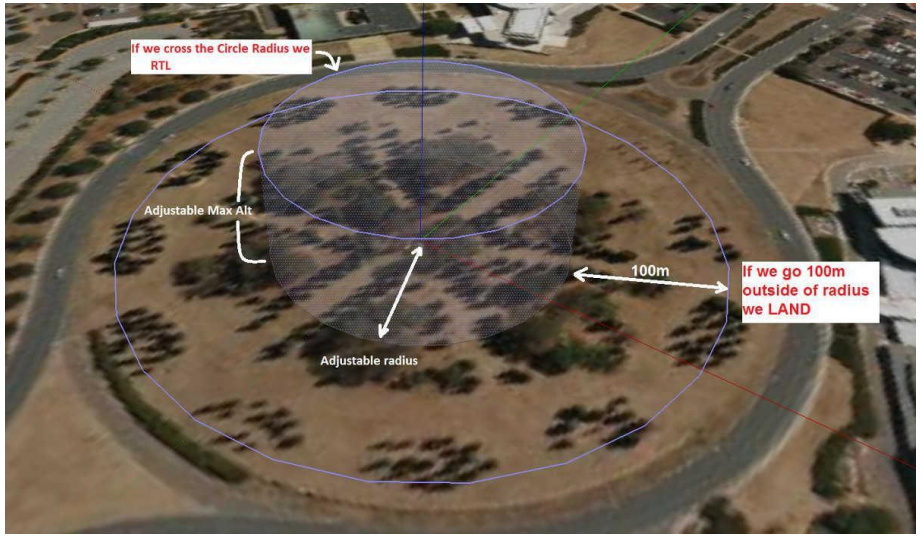
- Cameras
- Air measurements
- Robotic arms
- Any necessary equipment…



UAS operation key terms

Geofence

GPS tells drone to never go outside the GPS range



Fly-Away Protection

Marketing Term used for:

- Geofencing
- Auto-landing
- Return-to-home (back to take-off point)





UAS operation key terms

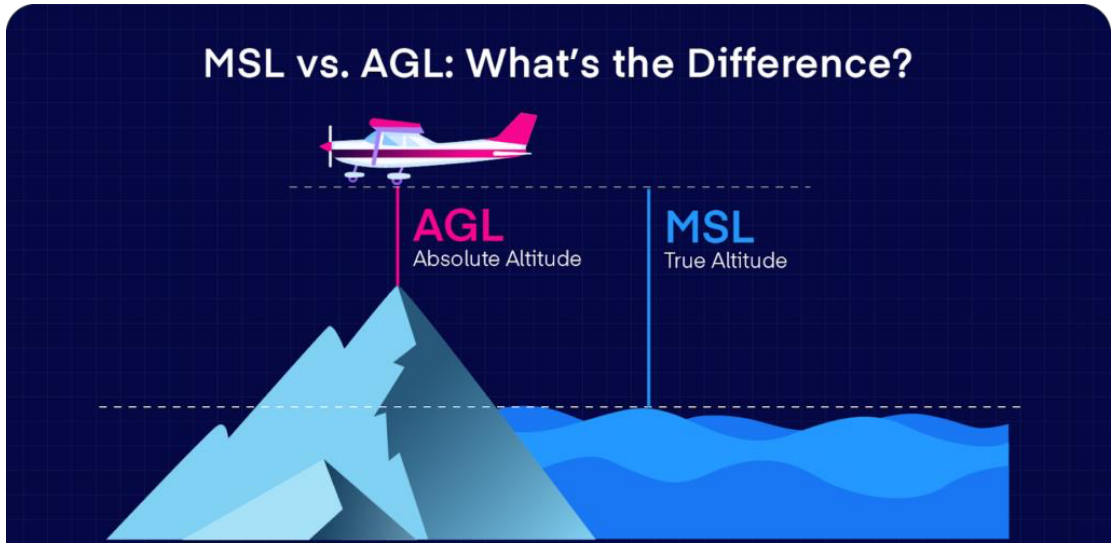
Optical Flow

Used to improve stability using a camera to detect the direction of motion



Above Ground Level (AGL)

- Indicates the drone's height from the terrain directly below it.
- Different from AMSL (above mean sea level).



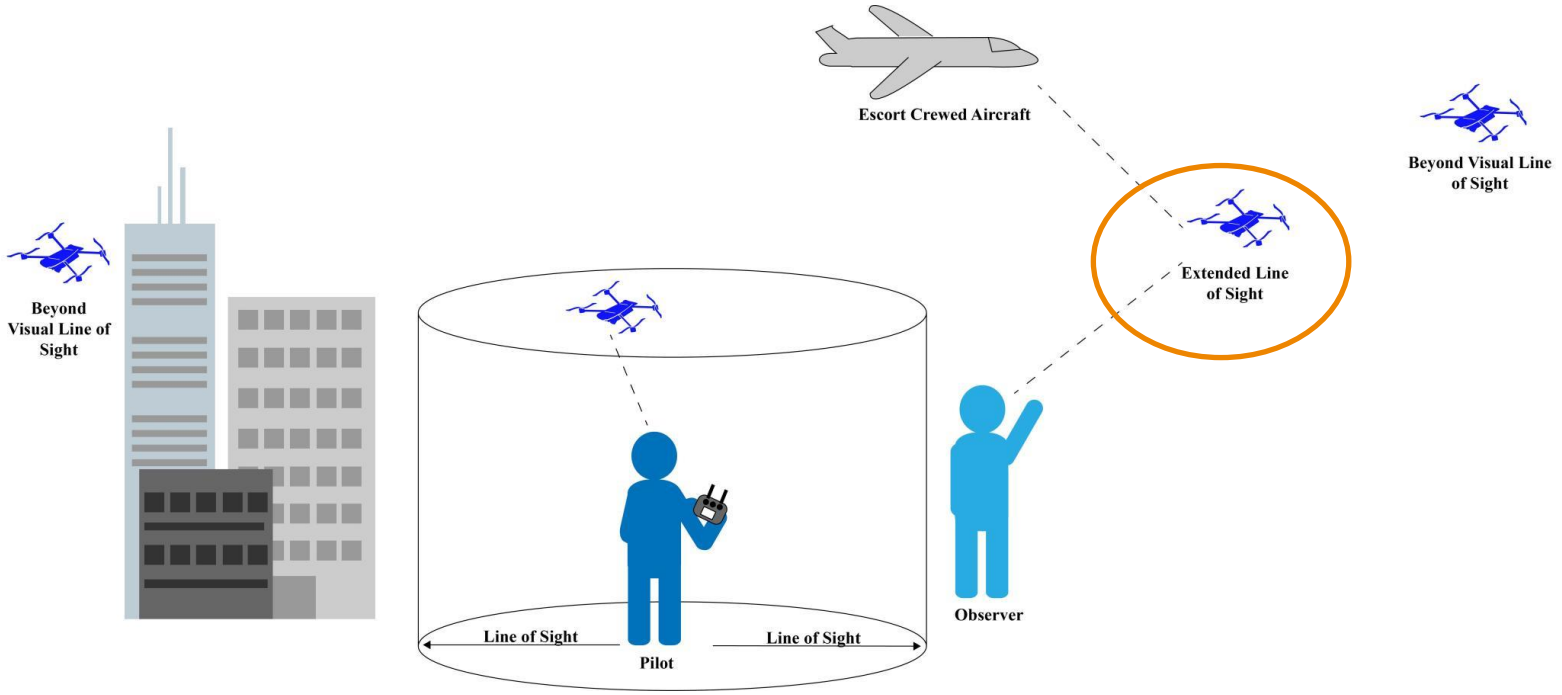
UAS operation key terms

Visual Line of Sight (VLOS)

- Straight line along which the observer has unobstructed view
- Abbreviated LOS

Beyond Visual Line of Sight (BVLOS)

- Operator and UAV too distant or obstructed
- Abbreviated BLOS



UAV crashes…

- Pilot Error
- Bad weather
- Loss of communication
- Fly-away (automated system error)
- Loss of GPS
- Loss of Altitude (IMU failure)





Thank You!